

JNVST Class 9 Question Bank Booklet

Real practice questions from the VedicNeev question bank, with full worked solutions.

Total Questions: 470 Easy: 176 Medium: 147 Hard: 147

Mental Ability (73 questions)

Q1. Find the missing number in the pattern: 1, 4, 9, 16, 25, ?

EASY

- A. 36 (Correct)**
- B. 25
- C. 35
- D. 30

Explanation: These are consecutive squares starting from 1^2 ($1^2, 2^2, 3^2, 4^2, 5^2$); the next is $6^2 = 36$.

Q2. Find the missing number in the pattern: 4, 9, 16, 25, 36, ?

EASY

- A. 36
- B. 48
- C. 42
- D. 49 (Correct)**

Explanation: These are consecutive squares starting from 2^2 ($2^2, 3^2, 4^2, 5^2, 6^2$); the next is $7^2 = 49$.

Q3. Find the missing number in the pattern: 9, 16, 25, 36, 49, ?

EASY

- A. 64 (Correct)**
- B. 49
- C. 63
- D. 56

Explanation: These are consecutive squares starting from 3^2 ($3^2, 4^2, 5^2, 6^2, 7^2$); the next is $8^2 = 64$.

Q4. Find the missing number in the pattern: 1, 4, 9, 16, 25, ?

EASY

- A. 36 (Correct)**
- B. 25
- C. 35
- D. 30

Explanation: These are consecutive squares starting from 1^2 ($1^2, 2^2, 3^2, 4^2, 5^2$); the next is $6^2 = 36$.

Q5. Find the missing number in the pattern: 16, 25, 36, 49, 64, ?

EASY

- A. 64
- B. 80
- C. 72
- D. 81 (Correct)**

Explanation: These are consecutive squares starting from 4^2 ($4^2, 5^2, 6^2, 7^2, 8^2$); the next is $9^2 = 81$.

Q6. Find the missing number in the pattern: 4, 9, 16, 25, 36, ?

EASY

- A. 36
- B. 48
- C. 42
- D. 49 (Correct)**

Explanation: These are consecutive squares starting from 2^2 ($2^2, 3^2, 4^2, 5^2, 6^2$); the next is $7^2 = 49$.

Q7. Find the missing number in the pattern: 25, 36, 49, 64, 81, ?

EASY

- A. 100 (Correct)**
- B. 81
- C. 99
- D. 90

Explanation: These are consecutive squares starting from 5^2 ($5^2, 6^2, 7^2, 8^2, 9^2$); the next is $10^2 = 100$.

Q8. Find the missing number in the pattern: 1, 8, 27, 64, 125, ?

EASY

- A. 216 (Correct)**
- B. 125
- C. 215
- D. 180

Explanation: These are consecutive cubes starting from 1^3 ($1^3, 2^3, 3^3, 4^3, 5^3$); the next is $6^3 = 216$.

Q9. Find the missing number in the pattern: 8, 27, 64, 125, 216, ?

EASY

- A. 294
- B. 343 (Correct)**
- C. 216
- D. 342

Explanation: These are consecutive cubes starting from 2^3 ($2^3, 3^3, 4^3, 5^3, 6^3$); the next is $7^3 = 343$.

Q10. Find the missing number in the pattern: 9, 16, 25, 36, 49, ?

EASY

- A. 64 (Correct)**
- B. 49
- C. 63
- D. 56

Explanation: These are consecutive squares starting from 3^2 ($3^2, 4^2, 5^2, 6^2, 7^2$); the next is $8^2 = 64$.

Q11. Find the missing number in the series: 3, 5, 7, 9, 11, ?

EASY

- A. 12
- B. 14
- C. 9
- D. 13 (Correct)**

Explanation: The series increases by a fixed common difference of 2 each time (3 '5 '7 '9 '11 '13). Continue the pattern: $11 + 2 = 13$.

Q12. Find the missing number in the series: 5, 8, 11, 14, 17, ?

EASY

- A. 20 (Correct)**
- B. 19
- C. 21
- D. 14

Explanation: The series increases by a fixed common difference of 3 each time (5 '8 '11 '14 '17 '20). Continue the pattern: $17 + 3 = 20$.

Q13. Find the missing number in the series: 7, 11, 15, 19, 23, ?

EASY

- A. 19
- B. 27 (Correct)**
- C. 26
- D. 28

Explanation: The series increases by a fixed common difference of 4 each time (7 '11 '15 '19 '23 '27). Continue the pattern: $23 + 4 = 27$.

Q14. Find the missing number in the series: 2, 7, 12, 17, 22, ?

EASY

- A. 17
- B. 27 (Correct)**
- C. 26
- D. 28

Explanation: The series increases by a fixed common difference of 5 each time (2 '7 '12 '17 '22 '27). Continue the pattern: $22 + 5 = 27$.

Q15. Find the missing number in the series: 6, 12, 18, 24, 30, ?

EASY

- A. 36 (Correct)**
- B. 35
- C. 37
- D. 24

Explanation: The series increases by a fixed common difference of 6 each time (6 '12 '18 '24 '30 '36). Continue the pattern: $30 + 6 = 36$.

Q16. Find the missing number in the series: 4, 11, 18, 25, 32, ?

EASY

- A. 25
- B. 39 (Correct)**
- C. 38
- D. 40

Explanation: The series increases by a fixed common difference of 7 each time (4 '11 '18 '25 '32 '39). Continue the pattern: $32 + 7 = 39$.

Q17. Find the missing number in the series: 9, 17, 25, 33, 41, ?

EASY

- A. 48
- B. 50
- C. 33
- D. 49 (Correct)**

Explanation: The series increases by a fixed common difference of 8 each time (9 '17 '25 '33 '41 '49). Continue the pattern: $41 + 8 = 49$.

Q18. Find the missing number in the series: 3, 12, 21, 30, 39, ?

EASY

- A. 48 (Correct)**
- B. 47
- C. 49
- D. 30

Explanation: The series increases by a fixed common difference of 9 each time (3 '12 '21 '30 '39 '48). Continue the pattern: $39 + 9 = 48$.

Q19. Find the missing number in the series: 8, 19, 30, 41, 52, ?

EASY

- A. 41
- B. 63 (Correct)**
- C. 62
- D. 64

Explanation: The series increases by a fixed common difference of 11 each time (8 '19 '30 '41 '52 '63). Continue the pattern: $52 + 11 = 63$.

Q20. Find the missing number in the series: 5, 17, 29, 41, 53, ?

EASY

- A. 64
- B. 66
- C. 41
- D. 65 (Correct)**

Explanation: The series increases by a fixed common difference of 12 each time (5 '17 '29 '41 '53 '65). Continue the pattern: 53 + 12 = 65.

Q21. Find the missing number: 2, 4, 8, 16, ?, 64

EASY

- A. 24
- B. 32 (Correct)**
- C. 48
- D. 36

Explanation: Each number is double the previous one: $2 \times 2 = 4$, $4 \times 2 = 8$, $8 \times 2 = 16$, $16 \times 2 = 32$.

Q22. Find the missing letter: A, C, E, G, ?

EASY

- A. H
- B. I (Correct)**
- C. J
- D. F

Explanation: The series skips one letter each time (+2): ACEGI.

Q23. Find the odd one out: Carrot, Apple, Mango, Banana

EASY

- A. Banana
- B. Carrot (Correct)**
- C. Apple
- D. Mango

Explanation: Apple, Mango, Banana are all fruits; Carrot is not.

Q24. Find the odd one out: Sphere, Triangle, Square, Circle

EASY

- A. Sphere (Correct)**
- B. Triangle
- C. Square
- D. Circle

Explanation: Triangle, Square, Circle are all flat (2D) shapes; Sphere is not.

Q25. Find the odd one out: January, Monday, Wednesday, Friday

EASY

- A. Monday
- B. Wednesday
- C. Friday
- D. January (Correct)**

Explanation: Monday, Wednesday, Friday are all days of the week; January is not.

Q26. Find the odd one out: 9, 2, 4, 8

EASY

- A. 4
- B. 8
- C. 9 (Correct)**
- D. 2

Explanation: 2, 4, 8 are all even numbers; 9 is not.

Q27. Find the odd one out: Sparrow, Cow, Horse, Goat

EASY

- A. Goat
- B. Sparrow (Correct)**
- C. Cow
- D. Horse

Explanation: Cow, Horse, Goat are all land mammals; Sparrow is not.

Q28. Find the odd one out: Cricket bat, Guitar, Violin, Flute

EASY

- A. Cricket bat (Correct)**
- B. Guitar
- C. Violin
- D. Flute

Explanation: Guitar, Violin, Flute are all musical instruments; Cricket bat is not.

Q29. Find the odd one out: Asia, India, Japan, Brazil

EASY

- A. India
- B. Japan
- C. Brazil
- D. Asia (Correct)**

Explanation: India, Japan, Brazil are all countries; Asia is not.

Q30. If CAT = 24 (using each letter's position in the alphabet, A=1, B=2, ...), then DOG = ?

MEDIUM

- A. 55
- B. 6
- C. 26 (Correct)**
- D. 27

Explanation: Each letter is coded by its position in the alphabet (A=1 ... Z=26) and the code is their sum. For DOG: D=4, O=15, G=7 'sum = 26.

Q31. If DOG = 26 (using each letter's position in the alphabet, A=1, B=2, ...), then CAT = ?

MEDIUM

- A. 24 (Correct)**
- B. 25
- C. 57
- D. 17

Explanation: Each letter is coded by its position in the alphabet (A=1 ... Z=26) and the code is their sum. For CAT: C=3, A=1, T=20 'sum = 24.

Q32. If SUN = 54 (using each letter's position in the alphabet, A=1, B=2, ...), then MAP = ?

MEDIUM

- A. 51
- B. 16
- C. 30 (Correct)**
- D. 31

Explanation: Each letter is coded by its position in the alphabet (A=1 ... Z=26) and the code is their sum. For MAP: M=13, A=1, P=16 'sum = 30.

Q33. If BAT = 23 (using each letter's position in the alphabet, A=1, B=2, ...), then SUN = ?

MEDIUM

- A. 27
- B. 34
- C. 54 (Correct)**
- D. 55

Explanation: Each letter is coded by its position in the alphabet (A=1 ... Z=26) and the code is their sum. For SUN: S=19, U=21, N=14 'sum = 54.

Q34. If MAP = 30 (using each letter's position in the alphabet, A=1, B=2, ...), then BAT = ?

MEDIUM

- A. 7
- B. 23 (Correct)**
- C. 24
- D. 58

*Explanation: Each letter is coded by its position in the alphabet (A=1 ... Z=26) and the code is their sum. For BAT: B=2, A=1, T=20
'sum' = 23.*

Q35. In each row below, the same rule turns the first two numbers into the third. Find the missing number:

Row 1: 2, 3, 5

Row 2: 4, 5, 19

Row 3: 6, 7, ?

MEDIUM

- A. 42
- B. 12
- C. 43
- D. 41 (Correct)**

Explanation: The rule is: third number = (first × second) + (first - second). Checking row 1: $2 \times 3 + (23) = 5$. Applying it to row 3: $6 \times 7 + (67) = 41$.

Q36. In each row below, the same rule turns the first two numbers into the third. Find the missing number:

Row 1: 3, 4, 11

Row 2: 5, 6, 29

Row 3: 7, 8, ?

MEDIUM

- A. 57
- B. 55 (Correct)**
- C. 56
- D. 14

Explanation: The rule is: third number = (first × second) + (first - second). Checking row 1: $3 \times 4 + (34) = 11$. Applying it to row 3: $7 \times 8 + (78) = 55$.

Q37. In each row below, the same rule turns the first two numbers into the third. Find the missing number:

Row 1: 2, 5, 7

Row 2: 3, 6, 15

Row 3: 4, 9, ?

MEDIUM

- A. 41
- B. 31 (Correct)**
- C. 36
- D. 8

Explanation: The rule is: third number = (first × second) + (first - second). Checking row 1: $2 \times 5 + (25) = 7$. Applying it to row 3: $4 \times 9 + (49) = 31$.

Q38. In each row below, the same rule turns the first two numbers into the third. Find the missing number:

Row 1: 1, 6, 1

Row 2: 2, 7, 9

Row 3: 3, 9, ?

MEDIUM

- A. 27
- B. 6
- C. 33
- D. 21 (Correct)**

Explanation: The rule is: third number = (first × second) + (first - second). Checking row 1: $1 \times 6 + (16) = 1$. Applying it to row 3: $3 \times 9 + (39) = 21$.

Q39. In each row below, the same rule turns the first two numbers into the third. Find the missing number:

Row 1: 4, 2, 10

Row 2: 6, 3, 21

Row 3: 8, 5, ?

MEDIUM

- A. 37
- B. 43 (Correct)**
- C. 40
- D. 16

Explanation: The rule is: third number = (first $\times$ second) + (first - second). Checking row 1: $4 \times 2 + (4 - 2) = 10$. Applying it to row 3: $8 \times 5 + (8 - 5) = 43$.

Q40. Find the missing number in the pattern: 1, 4, 9, 16, 25, ?

MEDIUM

- A. 30
- B. 36 (Correct)**
- C. 32
- D. 49

Explanation: These are perfect squares ($1^2, 2^2, 3^2, 4^2, 5^2, 6^2$); the next is $6^2 = 36$.

Q41. 7 : 14 :: 9 : ?

MEDIUM

- A. 17
- B. 20
- C. 18 (Correct)**
- D. 19

Explanation: The rule is: double the first number (7 $\times$ 14). Applying it to 9: double the 9 = 18.

Q42. 4 : 12 :: 6 : ?

MEDIUM

- A. 17
- B. 20
- C. 18 (Correct)**
- D. 19

Explanation: The rule is: triple the first number (4 $\times$ 12). Applying it to 6: triple the 6 = 18.

Q43. 10 : 15 :: 12 : ?

MEDIUM

- A. 18
- B. 16
- C. 19
- D. 17 (Correct)**

Explanation: The rule is: add 5 to the first number (10 $\times$ 15). Applying it to 12: add 5 to the 12 = 17.

Q44. 5 : 25 :: 7 : ?

MEDIUM

- A. 50
- B. 48
- C. 51
- D. 49 (Correct)**

Explanation: The rule is: square the first number (5 $\times$ 25). Applying it to 7: square the 7 = 49.

Q45. 3 : 11 :: 5 : ?

MEDIUM

- A. 21
- B. 19 (Correct)**
- C. 20
- D. 18

Explanation: The rule is: multiply the first number by 4 and subtract 1 (3 $\times$ 11). Applying it to 5: multiply the 5 by 4 and subtract 1 = 19.

Q46. Find the missing number in the pattern: 3, 9, 27, 81, ?

MEDIUM

- A. 162
- B. 202
- C. 243 (Correct)**
- D. 324

Explanation: Each number is multiplied by 3 to get the next: $81 \times 3 = 243$.

Q47. Find the missing number in the series: 10, 16, 14, 20, 18, ?

MEDIUM

- A. 24 (Correct)**
- B. 16
- C. 12
- D. 22

Explanation: The series alternates between adding 6 and subtracting 2 (+6, 2, +6, 2, ...). The last step was 2, so the next step adds 6: $18 + 6 = 24$.

Q48. Find the missing number in the series: 5, 13, 10, 18, 15, ?

MEDIUM

- A. 21
- B. 23 (Correct)**
- C. 12
- D. 7

Explanation: The series alternates between adding 8 and subtracting 3 (+8, 3, +8, 3, ...). The last step was 3, so the next step adds 8: $15 + 8 = 23$.

Q49. Find the missing number in the series: 20, 25, 16, 21, 12, ?

MEDIUM

- A. 3
- B. 7
- C. 19
- D. 17 (Correct)**

Explanation: The series alternates between adding 5 and subtracting 9 (+5, 9, +5, 9, ...). The last step was 9, so the next step adds 5: $12 + 5 = 17$.

Q50. Find the missing number in the series: 15, 22, 18, 25, 21, ?

MEDIUM

- A. 28 (Correct)**
- B. 17
- C. 14
- D. 27

Explanation: The series alternates between adding 7 and subtracting 4 (+7, 4, +7, 4, ...). The last step was 4, so the next step adds 7: $21 + 7 = 28$.

Q51. Find the missing number in the series: 8, 18, 15, 25, 22, ?

MEDIUM

- A. 32 (Correct)**
- B. 19
- C. 12
- D. 29

Explanation: The series alternates between adding 10 and subtracting 3 (+10, 3, +10, 3, ...). The last step was 3, so the next step adds 10: $22 + 10 = 32$.

Q52. 3 : 10 :: 5 : ?

HARD

- A. 27
- B. 11
- C. 26 (Correct)**
- D. 25

Explanation: The rule is "square the first number, then add 1" ($3^2 + 1 = 10 = 10$). Applying it to 5: $5^2 + 1 = 25 + 1 = 26$.

Q53. 4 : 17 :: 6 : ?

HARD

- A. 36
- B. 38
- C. 13
- D. 37 (Correct)**

Explanation: The rule is "square the first number, then add 1" ($4^2+1=17=17$). Applying it to 6: $6^2+1=36+1=37$.

Q54. 2 : 5 :: 7 : ?

HARD

- A. 51
- B. 15
- C. 50 (Correct)**
- D. 49

Explanation: The rule is "square the first number, then add 1" ($2^2+1=5=5$). Applying it to 7: $7^2+1=49+1=50$.

Q55. 5 : 26 :: 3 : ?

HARD

- A. 11
- B. 7
- C. 10 (Correct)**
- D. 9

Explanation: The rule is "square the first number, then add 1" ($5^2+1=26=26$). Applying it to 3: $3^2+1=9+1=10$.

Q56. 6 : 37 :: 4 : ?

HARD

- A. 16
- B. 18
- C. 9
- D. 17 (Correct)**

Explanation: The rule is "square the first number, then add 1" ($6^2+1=37=37$). Applying it to 4: $4^2+1=16+1=17$.

Q57. In a code, CAT is written as ECV (each letter shifted forward by a fixed number of places in the alphabet). Using the same rule, if the code for DOG is coded the same way and each letter's alphabet position (A=1...Z=26) is summed, what is that total?

HARD

- A. 32 (Correct)**
- B. 29
- C. 35
- D. 20

Explanation: CATECV shows a forward shift of 2 letters (e.g. CE). Applying the same +2 shift to DOG gives FQI, whose letter-position sum is 32.

Q58. In a code, DOG is written as GRJ (each letter shifted forward by a fixed number of places in the alphabet). Using the same rule, if the code for CAT is coded the same way and each letter's alphabet position (A=1...Z=26) is summed, what is that total?

HARD

- A. 30
- B. 36
- C. 67
- D. 33 (Correct)**

Explanation: DOGGRJ shows a forward shift of 3 letters (e.g. DG). Applying the same +3 shift to CAT gives FDW, whose letter-position sum is 33.

Q59. In a code, SUN is written as WYR (each letter shifted forward by a fixed number of places in the alphabet). Using the same rule, if the code for BAT is coded the same way and each letter's alphabet position (A=1...Z=26) is summed, what is that total?

HARD

- A. 63
- B. 35 (Correct)**
- C. 32
- D. 38

Explanation: SUNWYR shows a forward shift of 4 letters (e.g. SW). Applying the same +4 shift to BAT gives FEX, whose letter-position sum is 35.

Q60. In a code, MAP is written as RFU (each letter shifted forward by a fixed number of places in the alphabet). Using the same rule, if the code for SUN is coded the same way and each letter's alphabet position (A=1...Z=26) is summed, what is that total?

HARD

- A. 66
- B. 46
- C. 39
- D. 69 (Correct)**

Explanation: MAPRFU shows a forward shift of 5 letters (e.g. MR). Applying the same +5 shift to SUN gives XZS, whose letter-position sum is 69.

Q61. In a code, BAT is written as CBU (each letter shifted forward by a fixed number of places in the alphabet). Using the same rule, if the code for MAP is coded the same way and each letter's alphabet position (A=1...Z=26) is summed, what is that total?

HARD

- A. 30
- B. 36
- C. 53
- D. 33 (Correct)**

Explanation: BATCBU shows a forward shift of 1 letters (e.g. BC). Applying the same +1 shift to MAP gives NBQ, whose letter-position sum is 33.

Q62. In this 3×3 grid, every row adds up to the same total. Find the missing number: 4, 9, 2 | 3, 5, 7 | 8, 1, ?

HARD

- A. 8
- B. 9
- C. 6 (Correct)**
- D. 7

Explanation: Every row sums to 15 (row 1: 4+9+2=15; row 2: 3+5+7=15). So row 3's missing number is $15 - 8 - 1 = 6$.

Q63. In this 3×3 grid, every row adds up to the same total. Find the missing number: 6, 1, 8 | 7, 5, 3 | 2, 9, ?

HARD

- A. 4 (Correct)**
- B. 13
- C. 6
- D. 11

Explanation: Every row sums to 15 (row 1: 6+1+8=15; row 2: 7+5+3=15). So row 3's missing number is $15 - 2 - 9 = 4$.

Q64. In this 3×3 grid, every row adds up to the same total. Find the missing number: 2, 7, 6 | 9, 5, 1 | 4, 3, ?

HARD

- A. 8 (Correct)**
- B. 11
- C. 10
- D. 7

Explanation: Every row sums to 15 (row 1: 2+7+6=15; row 2: 9+5+1=15). So row 3's missing number is $15 - 4 - 3 = 8$.

Q65. In this 3×3 grid, every row adds up to the same total. Find the missing number: 8, 3, 4 | 1, 5, 9 | 6, 7, ?

HARD

- A. 4
- B. 13
- C. 2 (Correct)**
- D. 9

Explanation: Every row sums to 15 (row 1: $8+3+4=15$; row 2: $1+5+9=15$). So row 3's missing number is $15 - 6 - 7 = 2$.

Q66. In this 3×3 grid, every row adds up to the same total. Find the missing number: 1, 5, 9 | 8, 3, 4 | 6, 7, ?

HARD

- A. 4
- B. 13
- C. 2 (Correct)**
- D. 9

Explanation: Every row sums to 15 (row 1: $1+5+9=15$; row 2: $8+3+4=15$). So row 3's missing number is $15 - 6 - 7 = 2$.

Q67. Find the missing number in the series: 1, 1, 2, 3, 5, ?

HARD

- A. 8 (Correct)**
- B. 5
- C. 10
- D. 6

Explanation: Each term is the sum of the two terms before it ($1+1=2$, $1+2=3$, $2+3=5$), so the next term is $3+5=8$.

Q68. Find the missing number in the series: 2, 3, 5, 8, 13, ?

HARD

- A. 13
- B. 26
- C. 16
- D. 21 (Correct)**

Explanation: Each term is the sum of the two terms before it ($2+3=5$, $3+5=8$, $5+8=13$), so the next term is $8+13=21$.

Q69. Find the missing number in the series: 1, 4, 5, 9, 14, ?

HARD

- A. 18
- B. 23 (Correct)**
- C. 14
- D. 28

Explanation: Each term is the sum of the two terms before it ($1+4=5$, $4+5=9$, $5+9=14$), so the next term is $9+14=23$.

Q70. Find the missing number in the series: 3, 5, 8, 13, 21, ?

HARD

- A. 42
- B. 26
- C. 34 (Correct)**
- D. 21

Explanation: Each term is the sum of the two terms before it ($3+5=8$, $5+8=13$, $8+13=21$), so the next term is $13+21=34$.

Q71. Find the missing number in the series: 2, 2, 4, 6, 10, ?

HARD

- A. 16 (Correct)**
- B. 10
- C. 20
- D. 12

Explanation: Each term is the sum of the two terms before it ($2+2=4$, $2+4=6$, $4+6=10$), so the next term is $6+10=16$.

Q72. Find the missing number in the series: 1, 10, 3, 15, 5, ?

HARD

- A. 20 (Correct)**
- B. 7
- C. 17
- D. 10

Explanation: This is actually two interleaved series: the 1st/3rd/5th positions (1, 3, 5) rise by 2, and the 2nd/4th positions (10, 15) rise by 5. The missing 6th term continues the second series: $15 + 5 = 20$.

Q73. Find the missing number in the series: 3, 20, 6, 24, 9, ?

HARD

- A. 28 (Correct)**
- B. 12
- C. 27
- D. 13

Explanation: This is actually two interleaved series: the 1st/3rd/5th positions (3, 6, 9) rise by 3, and the 2nd/4th positions (20, 24) rise by 4. The missing 6th term continues the second series: $24 + 4 = 28$.

Arithmetic (73 questions)

Q1. Calculate quickly using the "all from 9, last from 10" shortcut: $100 - 63 = ?$

EASY

- A. 36
- B. 38
- C. 47
- D. 37 (Correct)**

Explanation: Subtract every digit of 63 from 9 except the last, which is subtracted from 10: $100 - 63 = 37$.

Q2. Calculate quickly using the "all from 9, last from 10" shortcut: $100 - 38 = ?$

EASY

- A. 63
- B. 72
- C. 62 (Correct)**
- D. 61

Explanation: Subtract every digit of 38 from 9 except the last, which is subtracted from 10: $100 - 38 = 62$.

Q3. Calculate quickly using vertically-and-crosswise multiplication: $12 \times 43 = ?$

EASY

- A. 516 (Correct)**
- B. 546
- C. 526
- D. 506

Explanation: Units: $2 \times 3 = 6$. Cross-multiply and add: $1 \times 3 + 2 \times 4 = 11$. Leading: $1 \times 4 = 4$. Combine with carries: $12 \times 43 = 516$.

Q4. Calculate quickly using vertically-and-crosswise multiplication: $21 \times 14 = ?$

EASY

- A. 304
- B. 284
- C. 294 (Correct)**
- D. 264

Explanation: Units: $1 \times 4 = 4$. Cross-multiply and add: $2 \times 4 + 1 \times 1 = 9$. Leading: $2 \times 1 = 2$. Combine with carries: $21 \times 14 = 294$.

Q5. Calculate quickly: $72 \times 11 = ?$

EASY

- A. 772
- B. 792 (Correct)**
- C. 702
- D. 812

Explanation: Sandwich rule for $\times 11$: 7_2 with the middle digit $7+2=9$ '792.

Q6. Calculate quickly using the Nikhilam near-base shortcut: $98 \times 97 = ?$

EASY

- A. 9500
- B. 10106
- C. 9506 (Correct)**
- D. 9494

Explanation: Both numbers are close to 100, with deviations -2 and -3. Cross-add one number and the other's deviation ($98-3 = 95$), multiply by the base, then add the product of the deviations ($-2 \times -3 = 6$): $100 \times 95 + 6 = 9506$.

Q7. Calculate quickly using the Nikhilam near-base shortcut: $99 \times 96 = ?$

EASY

- A. 9504 (Correct)**
- B. 9496
- C. 9500
- D. 10304

Explanation: Both numbers are close to 100, with deviations -1 and -4. Cross-add one number and the other's deviation ($99-4 = 95$), multiply by the base, then add the product of the deviations ($-1 \times -4 = 4$): $100 \times 95 + 4 = 9504$.

Q8. Calculate quickly using the "square ending in 5" shortcut: $15^2 = ?$

EASY

- A. 125
- B. 25
- C. 220
- D. 225 (Correct)**

Explanation: For a number $10a+5$ (here $a=1$), the square is $a \times (a+1)$ followed by 25: $1 \times 2 = 2$, so $15^2 = 225$.

Q9. Calculate quickly using the "square ending in 5" shortcut: $25^2 = ?$

EASY

- A. 425
- B. 225
- C. 620
- D. 625 (Correct)**

Explanation: For a number $10a+5$ (here $a=2$), the square is $a \times (a+1)$ followed by 25: $2 \times 3 = 6$, so $25^2 = 625$.

Q10. Calculate quickly using the $\times 11$ shortcut: $23 \times 11 = ?$

EASY

- A. 34
- B. 230
- C. 276
- D. 253 (Correct)**

Explanation: Sandwich the digit sum of 23 between its own digits (carrying if the sum reaches 10 or more): $23 \times 11 = 253$.

Q11. Calculate quickly using the $\times 11$ shortcut: $41 \times 11 = ?$

EASY

- A. 492
- B. 451 (Correct)**
- C. 52
- D. 410

Explanation: Sandwich the digit sum of 41 between its own digits (carrying if the sum reaches 10 or more): $41 \times 11 = 451$.

Q12. Calculate quickly: $45 \times 11 = ?$

EASY

- A. 485
- B. 495 (Correct)**
- C. 450
- D. 545

Explanation: Sandwich rule for $\times 11$: 4_5 with the middle digit $4+5=9$ '495.

Q13. What is the HCF (Highest Common Factor) of 12 and 18?

EASY

- A. 6 (Correct)**
- B. 8
- C. 12
- D. 4

Explanation: The factors of 12 are {1, 2, 3, 4, 6, 12} and of 18 are {1, 2, 3, 6, 9, 18}. The largest number common to both lists is 6.

Q14. What is the HCF (Highest Common Factor) of 24 and 36?

EASY

- A. 14
- B. 12 (Correct)**
- C. 24
- D. 10

Explanation: The factors of 24 are {1, 2, 3, 4, 6, 8, 12, 24} and of 36 are {1, 2, 3, 4, 6, 9, 12, 18, 36}. The largest number common to both lists is 12.

Q15. What is the HCF (Highest Common Factor) of 15 and 25?

EASY

- A. 7
- B. 15
- C. 5 (Correct)**
- D. 3

Explanation: The factors of 15 are {1, 3, 5, 15} and of 25 are {1, 5, 25}. The largest number common to both lists is 5.

Q16. What is the HCF (Highest Common Factor) of 8 and 20?

EASY

- A. 6
- B. 8
- C. 2
- D. 4 (Correct)**

Explanation: The factors of 8 are {1, 2, 4, 8} and of 20 are {1, 2, 4, 5, 10, 20}. The largest number common to both lists is 4.

Q17. What is the HCF (Highest Common Factor) of 14 and 21?

EASY

- A. 7 (Correct)**
- B. 9
- C. 14
- D. 5

Explanation: The factors of 14 are {1, 2, 7, 14} and of 21 are {1, 3, 7, 21}. The largest number common to both lists is 7.

Q18. What is the HCF (Highest Common Factor) of 16 and 40?

EASY

- A. 10
- B. 8 (Correct)**
- C. 16
- D. 6

Explanation: The factors of 16 are {1, 2, 4, 8, 16} and of 40 are {1, 2, 4, 5, 8, 10, 20, 40}. The largest number common to both lists is 8.

Q19. What is the HCF (Highest Common Factor) of 18 and 30?

EASY

- A. 8
- B. 18
- C. 6 (Correct)**
- D. 4

Explanation: The factors of 18 are {1, 2, 3, 6, 9, 18} and of 30 are {1, 2, 3, 5, 6, 10, 15, 30}. The largest number common to both lists is 6.

Q20. What is the HCF (Highest Common Factor) of 27 and 45?

EASY

- A. 11
- B. 27
- C. 7
- D. 9 (Correct)**

Explanation: The factors of 27 are {1, 3, 9, 27} and of 45 are {1, 3, 5, 9, 15, 45}. The largest number common to both lists is 9.

Q21. What is the HCF (Highest Common Factor) of 10 and 15?

EASY

- A. 5 (Correct)**
- B. 7
- C. 10
- D. 3

Explanation: The factors of 10 are {1, 2, 5, 10} and of 15 are {1, 3, 5, 15}. The largest number common to both lists is 5.

Q22. What is the HCF (Highest Common Factor) of 22 and 33?

EASY

- A. 13
- B. 11 (Correct)**
- C. 22
- D. 9

Explanation: The factors of 22 are {1, 2, 11, 22} and of 33 are {1, 3, 11, 33}. The largest number common to both lists is 11.

Q23. Calculate: $\frac{2}{7} + \frac{3}{7} = ?$

EASY

- A. $\frac{5}{7}$ (Correct)**
- B. $\frac{5}{14}$
- C. $-\frac{1}{7}$
- D. $\frac{6}{7}$

Explanation: The denominators are already the same, so just add the numerators: $2 + 3 = 5$, giving $\frac{5}{7}$, which simplifies to $\frac{5}{7}$.

Q24. Calculate: $\frac{5}{9} - \frac{2}{9} = ?$

EASY

- A. $\frac{7}{18}$
- B. $\frac{1}{3}$ (Correct)**
- C. $\frac{7}{9}$
- D. $\frac{2}{3}$

Explanation: The denominators are already the same, so just subtract the numerators: $5 - 2 = 3$, giving $\frac{3}{9}$, which simplifies to $\frac{1}{3}$.

Q25. Calculate: $\frac{1}{5} + \frac{2}{5} = ?$

EASY

- A. $\frac{3}{10}$
- B. $-\frac{1}{5}$
- C. $\frac{3}{5}$ (Correct)**
- D. $\frac{4}{5}$

Explanation: The denominators are already the same, so just add the numerators: $1 + 2 = 3$, giving $\frac{3}{5}$, which simplifies to $\frac{3}{5}$.

Q26. Calculate: $\frac{7}{10} - \frac{3}{10} = ?$

EASY

- A. $\frac{1}{2}$
- B. $\frac{1}{1}$
- C. $\frac{3}{5}$
- D. $\frac{2}{5}$ (Correct)**

Explanation: The denominators are already the same, so just subtract the numerators: $7 - 3 = 4$, giving $\frac{4}{10}$, which simplifies to $\frac{2}{5}$.

Q27. Calculate: $\frac{3}{8} + \frac{2}{8} = ?$

EASY

- A. $\frac{5}{8}$ (Correct)**
- B. $\frac{5}{16}$
- C. $\frac{1}{8}$
- D. $\frac{3}{4}$

Explanation: The denominators are already the same, so just add the numerators: $3 + 2 = 5$, giving $\frac{5}{8}$, which simplifies to $\frac{5}{8}$.

Q28. Calculate: $\frac{4}{11} + \frac{1}{11} = ?$

EASY

- A. $\frac{5}{22}$
- B. $\frac{5}{11}$ (Correct)**
- C. $\frac{3}{11}$
- D. $\frac{6}{11}$

Explanation: The denominators are already the same, so just add the numerators: $4 + 1 = 5$, giving $\frac{5}{11}$, which simplifies to $\frac{5}{11}$.

Q29. Calculate: $\frac{6}{13} - \frac{2}{13} = ?$

EASY

- A. $\frac{8}{13}$
- B. $\frac{5}{13}$
- C. $\frac{4}{13}$ (Correct)**
- D. $\frac{1}{13}$

Explanation: The denominators are already the same, so just subtract the numerators: $6 - 2 = 4$, giving $\frac{4}{13}$, which simplifies to $\frac{4}{13}$.

Q30. Calculate quickly using the "all from 9, last from 10" shortcut: $1000 - 587 = ?$

MEDIUM

- A. 412
- B. 414
- C. 423
- D. 413 (Correct)**

Explanation: Subtract every digit of 587 from 9 except the last, which is subtracted from 10: $1000 - 587 = 413$.

Q31. Calculate quickly using the "all from 9, last from 10" shortcut: $1000 - 234 = ?$

MEDIUM

- A. 767
- B. 776
- C. 766 (Correct)**
- D. 765

Explanation: Subtract every digit of 234 from 9 except the last, which is subtracted from 10: $1000 - 234 = 766$.

Q32. Calculate quickly using the "all from 9, last from 10" shortcut: $1000 - 456 = ?$

MEDIUM

- A. 544 (Correct)**
- B. 543
- C. 545
- D. 554

Explanation: Subtract every digit of 456 from 9 except the last, which is subtracted from 10: $1000 - 456 = 544$.

Q33. Calculate quickly using vertically-and-crosswise multiplication: $34 \times 56 = ?$

MEDIUM

- A. 1904 (Correct)**
- B. 1944
- C. 1914
- D. 1894

Explanation: Units: $4 \times 6 = 24$. Cross-multiply and add: $3 \times 6 + 4 \times 5 = 38$. Leading: $3 \times 5 = 15$. Combine with carries: $34 \times 56 = 1904$.

Q34. Calculate quickly using vertically-and-crosswise multiplication: $47 \times 68 = ?$

MEDIUM

- A. 3196 (Correct)**
- B. 3216
- C. 3206
- D. 3186

Explanation: Units: $7 \times 8 = 56$. Cross-multiply and add: $4 \times 8 + 7 \times 6 = 74$. Leading: $4 \times 6 = 24$. Combine with carries: $47 \times 68 = 3196$.

Q35. Calculate quickly using vertically-and-crosswise multiplication: $63 \times 42 = ?$

MEDIUM

- A. 2656
- B. 2636
- C. 2646 (Correct)**
- D. 2666

Explanation: Units: $3 \times 2 = 6$. Cross-multiply and add: $6 \times 2 + 3 \times 4 = 24$. Leading: $6 \times 4 = 24$. Combine with carries: $63 \times 42 = 2646$.

Q36. Calculate quickly using the Nikhilam near-base shortcut: $103 \times 104 = ?$

MEDIUM

- A. 10712 (Correct)**
- B. 10688
- C. 10700
- D. 9912

Explanation: Both numbers are close to 100, with deviations +3 and +4. Cross-add one number and the other's deviation ($103+4 = 107$), multiply by the base, then add the product of the deviations ($3 \times 4 = 12$): $100 \times 107 + 12 = 10712$.

Q37. Calculate quickly using the Nikhilam near-base shortcut: $92 \times 88 = ?$

MEDIUM

- A. 8096 (Correct)**
- B. 7904
- C. 8000
- D. 10496

Explanation: Both numbers are close to 100, with deviations -8 and -12. Cross-add one number and the other's deviation ($92-12 = 80$), multiply by the base, then add the product of the deviations ($-8 \times -12 = 96$): $100 \times 80 + 96 = 8096$.

Q38. Calculate quickly using the Nikhilam near-base shortcut: $107 \times 102 = ?$

MEDIUM

- A. 10900
- B. 10514
- C. 10914 (Correct)**
- D. 10886

Explanation: Both numbers are close to 100, with deviations +7 and +2. Cross-add one number and the other's deviation ($107+2 = 109$), multiply by the base, then add the product of the deviations ($7 \times 2 = 14$): $100 \times 109 + 14 = 10914$.

Q39. Calculate quickly using the "square ending in 5" shortcut: $65^2 = ?$

MEDIUM

- A. 3625
- B. 3025
- C. 4220
- D. 4225 (Correct)**

Explanation: For a number $10a+5$ (here $a=6$), the square is $a \times (a+1)$ followed by 25: $6 \times 7 = 42$, so $65^2 = 4225$.

Q40. Calculate quickly using the "square ending in 5" shortcut: $75^2 = ?$

MEDIUM

- A. 4925
- B. 4225
- C. 5620
- D. 5625 (Correct)**

Explanation: For a number $10a+5$ (here $a=7$), the square is $a \times (a+1)$ followed by 25: $7 \times 8 = 56$, so $75^2 = 5625$.

Q41. Calculate quickly using the "square ending in 5" shortcut: $85^2 = ?$

MEDIUM

- A. 6425
- B. 5625
- C. 7220
- D. 7225 (Correct)**

Explanation: For a number $10a+5$ (here $a=8$), the square is $a \times (a+1)$ followed by 25: $8 \times 9 = 72$, so $85^2 = 7225$.

Q42. Calculate quickly using the $\times 11$ shortcut: $68 \times 11 = ?$

MEDIUM

- A. 748 (Correct)**
- B. 79
- C. 680
- D. 816

Explanation: Sandwich the digit sum of 68 between its own digits (carrying if the sum reaches 10 or more): $68 \times 11 = 748$.

Q43. Calculate quickly using the $\times 11$ shortcut: $57 \times 11 = ?$

MEDIUM

- A. 684
- B. 627 (Correct)**
- C. 68
- D. 570

Explanation: Sandwich the digit sum of 57 between its own digits (carrying if the sum reaches 10 or more): $57 \times 11 = 627$.

Q44. Calculate quickly using the $\times 11$ shortcut: $75 \times 11 = ?$

MEDIUM

- A. 86
- B. 750
- C. 900
- D. 825 (Correct)**

Explanation: Sandwich the digit sum of 75 between its own digits (carrying if the sum reaches 10 or more): $75 \times 11 = 825$.

Q45. Calculate quickly: $65^2 = ?$

MEDIUM

- A. 4025
- B. 4125
- C. 4225 (Correct)**
- D. 4325

Explanation: For a number ending in 5: $6 \times (6+1) = 42$, then append 25 '4225.

Q46. Use the all-from-9-last-from-10 method: $1000 - 587 = ?$

MEDIUM

- A. 313
- B. 413 (Correct)**
- C. 423
- D. 513

Explanation: $9-5=4$, $9-8=1$, $10-7=3$ '413.

Q47. How many factors does 12 have (including 1 and 12 itself)?

MEDIUM

- A. 5
- B. 7
- C. 6 (Correct)**
- D. 8

Explanation: The factors of 12 are $\{1, 2, 3, 4, 6, 12\}$ — that's 6 in total.

Q48. How many factors does 18 have (including 1 and 18 itself)?

MEDIUM

- A. 5
- B. 7
- C. 8
- D. 6 (Correct)**

Explanation: The factors of 18 are {1, 2, 3, 6, 9, 18} — that's 6 in total.

Q49. How many factors does 24 have (including 1 and 24 itself)?

MEDIUM

- A. 8 (Correct)**
- B. 7
- C. 9
- D. 10

Explanation: The factors of 24 are {1, 2, 3, 4, 6, 8, 12, 24} — that's 8 in total.

Q50. How many factors does 20 have (including 1 and 20 itself)?

MEDIUM

- A. 5
- B. 6 (Correct)**
- C. 7
- D. 8

Explanation: The factors of 20 are {1, 2, 4, 5, 10, 20} — that's 6 in total.

Q51. How many factors does 28 have (including 1 and 28 itself)?

MEDIUM

- A. 5
- B. 7
- C. 6 (Correct)**
- D. 8

Explanation: The factors of 28 are {1, 2, 4, 7, 14, 28} — that's 6 in total.

Q52. Calculate quickly using the "all from 9, last from 10" shortcut: $10000 - 6789 = ?$

HARD

- A. 3221
- B. 3211 (Correct)**
- C. 3210
- D. 3212

Explanation: Subtract every digit of 6789 from 9 except the last, which is subtracted from 10: $10000 - 6789 = 3211$.

Q53. Calculate quickly using the "all from 9, last from 10" shortcut: $10000 - 4321 = ?$

HARD

- A. 5689
- B. 5679 (Correct)**
- C. 5678
- D. 5680

Explanation: Subtract every digit of 4321 from 9 except the last, which is subtracted from 10: $10000 - 4321 = 5679$.

Q54. Calculate quickly using the "all from 9, last from 10" shortcut: $10000 - 9876 = ?$

HARD

- A. 124 (Correct)**
- B. 123
- C. 125
- D. 134

Explanation: Subtract every digit of 9876 from 9 except the last, which is subtracted from 10: $10000 - 9876 = 124$.

Q55. Calculate quickly using vertically-and-crosswise multiplication: $89 \times 76 = ?$

HARD

- A. 6764 (Correct)**
- B. 6794
- C. 6774
- D. 6754

Explanation: Units: $9 \times 6 = 54$. Cross-multiply and add: $8 \times 6 + 9 \times 7 = 111$. Leading: $8 \times 7 = 56$. Combine with carries: $89 \times 76 = 6764$.

Q56. Calculate quickly using vertically-and-crosswise multiplication: $93 \times 87 = ?$

HARD

- A. 8081
- B. 8091 (Correct)**
- C. 8051
- D. 8101

Explanation: Units: $3 \times 7 = 21$. Cross-multiply and add: $9 \times 7 + 3 \times 8 = 87$. Leading: $9 \times 8 = 72$. Combine with carries: $93 \times 87 = 8091$.

Q57. Calculate quickly using vertically-and-crosswise multiplication: $98 \times 67 = ?$

HARD

- A. 6576
- B. 6556
- C. 6566 (Correct)**
- D. 6596

Explanation: Units: $8 \times 7 = 56$. Cross-multiply and add: $9 \times 7 + 8 \times 6 = 111$. Leading: $9 \times 6 = 54$. Combine with carries: $98 \times 67 = 6566$.

Q58. Calculate quickly using the Nikhilam near-base shortcut: $998 \times 997 = ?$

HARD

- A. 995000
- B. 1001006
- C. 995006 (Correct)**
- D. 994994

Explanation: Both numbers are close to 1000, with deviations -2 and -3. Cross-add one number and the other's deviation ($998 - 3 = 995$), multiply by the base, then add the product of the deviations ($-2 \times -3 = 6$): $1000 \times 995 + 6 = 995006$.

Q59. Calculate quickly using the Nikhilam near-base shortcut: $1003 \times 1008 = ?$

HARD

- A. 1011024 (Correct)**
- B. 1010976
- C. 1011000
- D. 995024

Explanation: Both numbers are close to 1000, with deviations +3 and +8. Cross-add one number and the other's deviation ($1003 + 8 = 1011$), multiply by the base, then add the product of the deviations ($3 \times 8 = 24$): $1000 \times 1011 + 24 = 1011024$.

Q60. Calculate quickly using the Nikhilam near-base shortcut: $994 \times 991 = ?$

HARD

- A. 985000
- B. 1003054
- C. 985054 (Correct)**
- D. 984946

Explanation: Both numbers are close to 1000, with deviations -6 and -9. Cross-add one number and the other's deviation ($994 - 9 = 985$), multiply by the base, then add the product of the deviations ($-6 \times -9 = 54$): $1000 \times 985 + 54 = 985054$.

Q61. Calculate quickly using the "square ending in 5" shortcut: $105^2 = ?$

HARD

- A. 10025
- B. 9025
- C. 11020
- D. 11025 (Correct)**

Explanation: For a number $10a+5$ (here $a=10$), the square is $a \times (a+1)$ followed by 25: $10 \times 11 = 110$, so $105^2 = 11025$.

Q62. Calculate quickly using the "square ending in 5" shortcut: $115^2 = ?$

HARD

- A. 12125
- B. 11025
- C. 13220
- D. 13225 (Correct)**

Explanation: For a number $10a+5$ (here $a=11$), the square is $a \times (a+1)$ followed by 25: $11 \times 12 = 132$, so $115^2 = 13225$.

Q63. Calculate quickly using the "square ending in 5" shortcut: $125^2 = ?$

HARD

- A. 14425
- B. 13225
- C. 15620
- D. 15625 (Correct)**

Explanation: For a number $10a+5$ (here $a=12$), the square is $a \times (a+1)$ followed by 25: $12 \times 13 = 156$, so $125^2 = 15625$.

Q64. Calculate quickly using the $\times 11$ shortcut: $234 \times 11 = ?$

HARD

- A. 2340
- B. 2808
- C. 2574 (Correct)**
- D. 245

Explanation: Sandwich the digit sum of 234 between its own digits (carrying if the sum reaches 10 or more): $234 \times 11 = 2574$.

Q65. Calculate quickly using the $\times 11$ shortcut: $456 \times 11 = ?$

HARD

- A. 5016 (Correct)**
- B. 467
- C. 4560
- D. 5472

Explanation: Sandwich the digit sum of 456 between its own digits (carrying if the sum reaches 10 or more): $456 \times 11 = 5016$.

Q66. Calculate quickly using the $\times 11$ shortcut: $123 \times 11 = ?$

HARD

- A. 134
- B. 1230
- C. 1476
- D. 1353 (Correct)**

Explanation: Sandwich the digit sum of 123 between its own digits (carrying if the sum reaches 10 or more): $123 \times 11 = 1353$.

Q67. Calculate quickly: $98 \times 97 = ?$

HARD

- A. 9406
- B. 9506 (Correct)**
- C. 9606
- D. 9516

Explanation: Base 100, deviations -2 and -3: $(98-3) \times 100 + (-2 \times -3) = 9500 + 6 = 9506$.

Q68. Calculate quickly: $102 \times 104 = ?$

HARD

- A. 10408
- B. 10508
- C. 10608 (Correct)**
- D. 10708

Explanation: Base 100, deviations +2 and +4: $(102+4) \times 100 + (2 \times 4) = 10600 + 8 = 10608$.

Q69. Three bells ring at intervals of 4, 6, and 8 minutes respectively. If they all ring together now, after how many minutes will they next ring together?

HARD

- A. 18
- B. 12
- C. 24 (Correct)**
- D. 16

Explanation: The bells ring together again after the LCM of their intervals: $LCM(4, 6, 8) = 24$ minutes.

Q70. Three bells ring at intervals of 5, 10, and 15 minutes respectively. If they all ring together now, after how many minutes will they next ring together?

HARD

- A. 10
- B. 40
- C. 15
- D. 30 (Correct)**

Explanation: The bells ring together again after the LCM of their intervals: $LCM(5, 10, 15) = 30$ minutes.

Q71. Three bells ring at intervals of 6, 9, and 12 minutes respectively. If they all ring together now, after how many minutes will they next ring together?

HARD

- A. 36 (Correct)**
- B. 27
- C. 18
- D. 24

Explanation: The bells ring together again after the LCM of their intervals: $LCM(6, 9, 12) = 36$ minutes.

Q72. Three bells ring at intervals of 3, 4, and 6 minutes respectively. If they all ring together now, after how many minutes will they next ring together?

HARD

- A. 13
- B. 12 (Correct)**
- C. 18
- D. 3

Explanation: The bells ring together again after the LCM of their intervals: $LCM(3, 4, 6) = 12$ minutes.

Q73. Three bells ring at intervals of 8, 12, and 16 minutes respectively. If they all ring together now, after how many minutes will they next ring together?

HARD

- A. 36
- B. 24
- C. 48 (Correct)**
- D. 32

Explanation: The bells ring together again after the LCM of their intervals: $LCM(8, 12, 16) = 48$ minutes.

Language (64 questions)

Q1. Which word in the sentence is a noun?
"The teacher wrote on the board."

EASY

- A. teacher (Correct)**
- B. wrote
- C. on
- D. the

Explanation: A noun is a naming word for a person, place, or thing. 'Teacher' names a person, so it is the noun in this sentence.

Q2. Choose the correct word to complete the sentence: I saw ____ elephant at the zoo yesterday.

EASY

- A. a
- B. an (Correct)**
- C. the
- D. (no article needed)

Explanation: 'Elephant' begins with a vowel sound, so the indefinite article 'an' is used. Since this is the first time the elephant is mentioned, the indefinite article is required, not 'the'.

Q3. Choose the correct verb to complete the sentence: She ____ to school every day.

EASY

- A. go
- B. goes (Correct)**
- C. going
- D. gone

Explanation: In the simple present tense, third-person singular subjects like 'she' take a verb ending in -s/-es. 'Goes' correctly agrees with 'she'.

Q4. Choose the grammatically correct sentence.

EASY

- A. The dogs barks loudly.
- B. The dogs bark loudly. (Correct)**
- C. The dog bark loudly.
- D. The dog barking loudly.

Explanation: The plural subject 'dogs' requires the base form of the verb without -s in the simple present tense, so 'bark' is correct.

Q5. Choose the correct preposition: I was born ____ 2014.

EASY

- A. in (Correct)**
- B. on
- C. at
- D. by

Explanation: The preposition 'in' is used with years, months, and longer periods of time, so 'in 2014' is correct.

Q6. Choose the synonym of 'happy'.

EASY

- A. sad
- B. joyful (Correct)**
- C. angry
- D. tired

Explanation: 'Joyful' means feeling great pleasure, which matches the meaning of 'happy'.

Q7. Choose the antonym of 'brave'.

EASY

- A. courageous
- B. bold
- C. cowardly (Correct)**
- D. strong

Explanation: 'Cowardly' means lacking courage, which is the direct opposite of 'brave'.

Q8. Choose the correctly spelled word.

EASY

- A. recieve
- B. receive (Correct)**
- C. receeve
- D. receve

Explanation: The correct spelling follows the pattern 'e' before 'i' after the letter 'c', giving 'receive'.

Q9. Which punctuation mark should end this sentence?

"What is your name"

EASY

- A. Full stop (.)
- B. Question mark (?) (Correct)**
- C. Exclamation mark (!)
- D. Comma (,)

Explanation: This sentence directly asks something, so it must end with a question mark.

Q10. Which word in the sentence is a verb?

"The bird sings sweetly in the morning."

EASY

- A. bird
- B. sings (Correct)**
- C. sweetly
- D. morning

Explanation: A verb shows an action or state. 'Sings' shows the action the bird performs, making it the verb.

Q11. Choose the correct passive voice form of the sentence: 'The gardener waters the plants every morning.'

EASY

- A. The plants are watered by the gardener every morning. (Correct)**
- B. The plants were watered by the gardener every morning.
- C. The plants is watered by the gardener every morning.
- D. The plants are watering by the gardener every morning.

Explanation: The original sentence is in the simple present tense with the plural object 'the plants' becoming the subject of the passive sentence. The passive structure needed is 'am/is/are + past participle'; since 'the plants' is plural, 'are watered' is correct.

Q12. Read the passage and answer the question:

"Meera visited the old library every Saturday afternoon. She loved the smell of ageing paper and the quiet that settled over the tall wooden shelves. Working meticulously, she catalogued each returned book before the librarian could even ask. Her quiet dedication surprised even her own teachers."

What is the main idea of the passage?

EASY

- A. Meera dislikes visiting the library.
- B. Meera shows quiet, careful dedication to her library work. (Correct)**
- C. Meera is a careless and disorganised visitor.
- D. The librarian scolds Meera every Saturday.

Explanation: The passage describes Meera loving the library's atmosphere and working 'meticulously' to catalogue books without being asked, which shows quiet, careful dedication — this is the passage's central idea.

Q13. Convert into indirect (reported) speech: He said, 'I am going to Delhi tomorrow.'

EASY

- A. He said that he is going to Delhi tomorrow.
- B. He said that he was going to Delhi tomorrow.
- C. He said that he was going to Delhi the next day. (Correct)**
- D. He said that he had gone to Delhi the next day.

Explanation: Since the reporting verb 'said' is in the past tense, the present continuous 'am going' backshifts to past continuous 'was going', and the time word 'tomorrow' changes to 'the next day' because the speech is now being reported later.

Q14. Read the passage and answer the question:

"Meera visited the old library every Saturday afternoon. She loved the smell of ageing paper and the quiet that settled over the tall wooden shelves. Working meticulously, she catalogued each returned book before the librarian could even ask. Her quiet dedication surprised even her own teachers."

In the passage, the word 'meticulously' is closest in meaning to:

EASY

- A. carelessly
- B. quickly and without thought
- C. unwillingly
- D. very carefully and with attention to detail (Correct)**

Explanation: 'Meticulously' describes doing something with great care and close attention to detail, which fits Meera cataloguing books precisely before being asked.

Q15. Choose the correct modal to complete the sentence: 'You ____ wear a helmet while riding a bike; it is the law.'

EASY

- A. must (Correct)**
- B. might
- C. may
- D. would

Explanation: 'Must' expresses strong necessity or a legal obligation, which matches the meaning that wearing a helmet is required by law.

Q16. Read the poem and answer the question:

"The wind sang softly through the trees,
Its breath as cool as morning dew.
It danced across the golden fields,
And whispered secrets no one knew."

Which line from the poem contains a simile?

EASY

- A. "The wind sang softly through the trees"
- B. "Its breath as cool as morning dew" (Correct)**
- C. "It danced across the golden fields"
- D. "And whispered secrets no one knew"

Explanation: A simile directly compares two unlike things using 'as' or 'like'. 'Its breath as cool as morning dew' uses the 'as...as' construction to compare the wind's coolness to morning dew.

Q17. Choose the sentence with the correct article: 'He is ____ honest man, and everyone trusts him.'

EASY

- A. a
- B. the
- C. an (Correct)**
- D. (no article needed)

Explanation: The 'h' in 'honest' is silent, so the word begins with a vowel sound. The indefinite article 'an' is used before words starting with a vowel sound, so 'an honest man' is correct.

Q18. Read the poem and answer the question:

"The wind sang softly through the trees,
Its breath as cool as morning dew.
It danced across the golden fields,
And whispered secrets no one knew."

Which poetic device is most clearly used throughout the poem?

EASY

- A. simile
- B. hyperbole
- C. alliteration
- D. personification (Correct)**

Explanation: Throughout the poem, the wind is given human actions — it 'sings', 'dances', and 'whispers' — which is the definition of personification: giving human qualities to a non-human thing.

Q19. Choose the correct form to complete the sentence: 'She ____ for three hours; she must be tired.'

EASY

- A. has been studying (Correct)**
- B. had studied
- C. is studying
- D. studies

Explanation: The present perfect continuous ('has been studying') is used for an action that began in the past and continues up to now, emphasising its duration ('for three hours') — exactly the meaning here.

Q20. Read the passage and answer the question:

"The village had not seen rain for three long months. The wells ran dry, and the once-green fields turned to dust. Farmers watched helplessly as their crops withered under the harsh sun. Many families began rationing water to make it last."

According to the passage, why did the farmers' crops wither?

EASY

- A. The farmers refused to water them.
- B. Prolonged lack of rain dried up the wells and fields. (Correct)**
- C. The farmers planted the wrong crops.
- D. A flood destroyed the fields.

Explanation: The passage states the village 'had not seen rain for three long months', which caused the wells to run dry and fields to turn to dust — this prolonged drought is the direct cause of the withered crops.

Q21. Choose the grammatically correct sentence.

MEDIUM

- A. While I was cooking, the phone rang. (Correct)**
- B. While I cooking, the phone rang.
- C. While I was cook, the phone rang.
- D. While I am cooking, the phone rang.

Explanation: An ongoing past action interrupted by a shorter past action uses past continuous ('was cooking') alongside simple past ('rang').

Q22. Choose the grammatically correct sentence.

MEDIUM

- A. Everest is highest mountain in the world.
- B. Everest is the highest mountain in the world. (Correct)**
- C. Everest is a highest mountain in the world.
- D. Everest is an highest mountain in the world.

Explanation: Superlative adjectives naming a unique, one-of-a-kind item always take the definite article 'the'.

Q23. Choose the grammatically correct sentence.

MEDIUM

- A. The furniture in this room are old.
- B. The furniture in this room is old. (Correct)**
- C. The furnitures in this room are old.
- D. The furniture in this room were old.

Explanation: 'Furniture' is an uncountable noun and always takes a singular verb, so 'is' is correct here.

Q24. Choose the correct preposition: I have lived in this town ____ ten years.

MEDIUM

- A. since
- B. for (Correct)**
- C. from
- D. at

Explanation: 'For' is used with a duration or length of time, such as 'ten years', so it correctly completes this sentence.

Q25. Choose the synonym of 'enormous'.

MEDIUM

- A. tiny
- B. huge (Correct)**
- C. narrow
- D. quiet

Explanation: 'Huge' means extremely large in size, which matches the meaning of 'enormous'.

Q26. Choose the antonym of 'ancient'.

MEDIUM

- A. old
- B. modern (Correct)**
- C. historic
- D. antique

Explanation: 'Modern' means belonging to the present time, which is the direct opposite of 'ancient'.

Q27. Choose the grammatically correct sentence.

MEDIUM

- A. She don't like mangoes.
- B. She doesn't like mangoes. (Correct)**
- C. She not like mangoes.
- D. She doesn't likes mangoes.

Explanation: Negative simple present sentences with third-person singular subjects use 'doesn't' plus the base form of the verb.

Q28. Choose the correctly punctuated sentence.

MEDIUM

- A. I bought apples oranges and bananas.
- B. I bought apples, oranges, and bananas. (Correct)**
- C. I bought, apples oranges and bananas.
- D. I bought apples oranges, and bananas.

Explanation: Items in a list are separated by commas, with a comma also placed before 'and' preceding the final item.

Q29. Choose the grammatically correct sentence.

MEDIUM

- A. He runs quick.
- B. He runs quickly. (Correct)**
- C. He is running quick.
- D. He runs quickest.

Explanation: A verb like 'runs' must be modified by an adverb, and 'quickly' is the correct adverb form of 'quick'.

Q30. Choose the grammatically correct sentence.

MEDIUM

- A. I have seen that movie yesterday.
- B. I saw that movie yesterday. (Correct)**
- C. I have saw that movie yesterday.
- D. I seen that movie yesterday.

Explanation: A specific past time marker like 'yesterday' requires the simple past tense, so 'saw' is correct.

Q31. Choose the correct word: Please ____ your seatbelt before the car starts.

MEDIUM

- A. where
- B. wear (Correct)**
- C. were
- D. we're

Explanation: 'Wear' is the verb meaning to put on or fasten something, which fits the instruction about the seatbelt.

Q32. Choose the grammatically correct sentence.

MEDIUM

- A. Everyone have finished their homework.
- B. Everyone has finished their homework. (Correct)**
- C. Everyone has finished they homework.
- D. Everyone has finish their homework.

Explanation: 'Everyone' is a singular indefinite pronoun and takes the singular verb 'has', with the possessive 'their' before the noun 'homework'.

Q33. Choose the correct preposition: She is good ____ solving puzzles.

MEDIUM

- A. in
- B. at (Correct)**
- C. on
- D. with

Explanation: 'Good at' is the fixed expression used to describe skill or ability at an activity like solving puzzles.

Q34. Which word in the sentence is a conjunction?

"I wanted to play, but it started raining."

MEDIUM

- A. wanted
- B. but (Correct)**
- C. started
- D. raining

Explanation: A conjunction joins two clauses. 'But' connects the clause about wanting to play with the clause about it raining.

Q35. Choose the grammatically correct sentence.

MEDIUM

- A. Yesterday, I go to the market and bought vegetables.
- B. Yesterday, I went to the market and bought vegetables. (Correct)**
- C. Yesterday, I went to the market and buy vegetables.
- D. Yesterday, I going to the market and bought vegetables.

Explanation: Both actions happened at the same past time marked by 'yesterday', so both verbs must be in the simple past tense: 'went' and 'bought'.

Q36. Choose the correct plural form of 'Child':

MEDIUM

- A. Childs
- B. Childes
- C. Children (Correct)**
- D. Childrens

Explanation: 'Child' has an irregular plural: 'Children', not '-s' or '-es'.

Q37. Change into passive voice: 'You must complete the assignment before Friday.'

MEDIUM

- A. The assignment must complete before Friday.
- B. The assignment must be completing before Friday.
- C. The assignment must be completed before Friday. (Correct)**
- D. The assignment must have completed before Friday.

Explanation: Passive voice with a modal follows the pattern 'modal + be + past participle'. Here that gives 'must be completed', correctly turning 'the assignment' into the subject.

Q38. Read the passage and answer the question:

"Arjun had trained for two years before he finally stood at the base of the towering peak. Each step upward tested muscles he did not know he had, yet he refused to turn back. When he reached the summit at dawn, an overwhelming stillness settled over him, quieter than anything he had felt before. He realised that the mountain had taught him more about patience than any classroom ever could."

What can be inferred about Arjun's character from the passage?

MEDIUM

- A. He gives up easily when faced with difficulty.
- B. He regrets attempting to climb the mountain.
- C. He is indifferent to the experience of climbing.
- D. He is persistent and values what hardship teaches him. (Correct)**

Explanation: Arjun trains for two years, 'refuses to turn back' despite the difficulty, and reflects that the climb taught him patience — this shows persistence and a deep appreciation for what the hardship taught him.

Q39. Convert to indirect speech: She asked, 'Where do you live?'

MEDIUM

- A. She asked where I lived. (Correct)**
- B. She asked where do I live?
- C. She asked that where I lived.
- D. She asked where I live.

Explanation: Indirect wh-questions use normal statement word order (subject before verb), drop the question mark, and backshift the tense since the reporting verb 'asked' is past; 'where I lived' correctly does all three.

Q40. Read the passage and answer the question:

"Arjun had trained for two years before he finally stood at the base of the towering peak. Each step upward tested muscles he did not know he had, yet he refused to turn back. When he reached the summit at dawn, an overwhelming stillness settled over him, quieter than anything he had felt before. He realised that the mountain had taught him more about patience than any classroom ever could."

The overall tone of the passage is best described as:

MEDIUM

- A. humorous and light-hearted
- B. reflective and admiring (Correct)**
- C. angry and bitter
- D. fearful and anxious

Explanation: The passage calmly reflects on Arjun's effort and the lesson in patience the mountain gave him, expressed with quiet admiration for the experience — this is a reflective, admiring tone.

Q41. Identify the noun clause in the sentence: 'I believe that honesty is the best policy.'

MEDIUM

- A. I believe
- B. honesty is the best
- C. that honesty is the best policy (Correct)**
- D. the best policy

Explanation: 'That honesty is the best policy' has its own subject ('honesty') and verb ('is') and functions as the object of the verb 'believe' — this makes it a noun clause.

Q42. Read the passage and answer the question:

"Arjun had trained for two years before he finally stood at the base of the towering peak. Each step upward tested muscles he did not know he had, yet he refused to turn back. When he reached the summit at dawn, an overwhelming stillness settled over him, quieter than anything he had felt before. He realised that the mountain had taught him more about patience than any classroom ever could."

In the passage, 'overwhelming' most nearly means:

MEDIUM

- A. very mild and barely noticeable
- B. confusing and unclear
- C. brief and forgettable
- D. extremely intense, almost too much to bear (Correct)**

Explanation: 'Overwhelming' describes something so intense that it is difficult to fully process, matching the description of a stillness 'quieter than anything he had felt before'.

Q43. Choose the grammatically correct sentence.

HARD

- A. By the time the guests arrived, she had cooked the meal. (Correct)**
- B. By the time the guests arrived, she has cooked the meal.
- C. By the time the guests arrived, she cooked the meal.
- D. By the time the guests arrived, she have cooked the meal.

Explanation: When one past action finished before another past action, the earlier action uses the past perfect tense ('had cooked'), while the later one uses simple past ('arrived').

Q44. Choose the grammatically correct sentence.

HARD

- A. Neither the teacher nor the students was late.
- B. Neither the teacher nor the students were late. (Correct)**
- C. Neither the teacher nor the students is late.
- D. Neither the teacher nor the students be late.

Explanation: With 'neither...nor', the verb agrees with the subject nearer to it. Since 'students' (plural) is nearer, the plural past-tense verb 'were' is correct.

Q45. Choose the grammatically correct sentence.

HARD

- A. The honesty is the best policy.
- B. Honesty is the best policy. (Correct)**
- C. An honesty is the best policy.
- D. A honesty is the best policy.

Explanation: Abstract, uncountable nouns used in a general sense, like 'honesty' here, are used without any article.

Q46. Choose the correct preposition: The students are afraid ____ the dark.

HARD

- A. of (Correct)**
- B. from
- C. with
- D. about

Explanation: 'Afraid of' is the standard fixed expression used to describe fear of something, such as the dark.

Q47. Choose the synonym of 'diligent'.

HARD

- A. lazy
- B. hardworking (Correct)**
- C. careless
- D. forgetful

Explanation: 'Diligent' means showing careful and persistent effort in work, which matches the meaning of 'hardworking'.

Q48. Choose the antonym of 'generous'.

HARD

- A. kind
- B. selfish (Correct)**
- C. wealthy
- D. polite

Explanation: 'Selfish' means concerned only with one's own interest and unwilling to share, which is the direct opposite of 'generous'.

Q49. Choose the grammatically correct sentence.

HARD

- A. Walking to school, the rain started falling on Ravi.
- B. Walking to school, Ravi got caught in the rain. (Correct)**
- C. Walking to school, the school bag got wet.
- D. Walking to school, it started to rain on Ravi's bag.

Explanation: The introductory phrase 'Walking to school' must be immediately followed by the person actually doing the walking, which is 'Ravi'.

Q50. Choose the correctly punctuated sentence.

HARD

- A. The dog's bone was buried in the yard. (Correct)**
- B. The dogs bone was buried in the yard.
- C. The dog's bones was buried in the yard.
- D. The dogs' bone were buried in the yard.

Explanation: An apostrophe followed by 's' shows singular possession, so "dog's bone" correctly shows the bone belonging to one dog, matching the singular verb 'was'.

Q51. Choose the grammatically correct sentence.

HARD

- A. This book is mine. (Correct)**
- B. This book is my.
- C. This book is me.
- D. This book is mines.

Explanation: 'Mine' is the possessive pronoun that can stand alone as a complement to show ownership, so it correctly completes this sentence.

Q52. Choose the grammatically correct sentence.

HARD

- A. By next year, I will complete my project.
- B. By next year, I will have completed my project. (Correct)**
- C. By next year, I will completing my project.
- D. By next year, I have completed my project.

Explanation: The future perfect tense ('will have completed') is used to show an action that will be finished before a specific point in the future, marked here by 'by next year'.

Q53. Choose the grammatically correct sentence.

HARD

- A. The basket of apples are on the table.
- B. The basket of apples is on the table. (Correct)**
- C. The baskets of apples is on the table.
- D. The basket of apple are on the table.

Explanation: The true subject of the sentence is the singular noun 'basket', not 'apples' inside the prepositional phrase, so the singular verb 'is' is correct.

Q54. Choose the correct word: ____ going to the park after lunch.

HARD

- A. Their
- B. There
- C. They're (Correct)**
- D. Thier

Explanation: 'They're' is the contraction of 'they are', which correctly fits before the verb phrase 'going to the park'.

Q55. Choose the grammatically correct sentence.

HARD

- A. She likes reading, swimming, and to dance.
- B. She likes reading, swimming, and dancing. (Correct)**
- C. She likes to read, swimming, and dancing.
- D. She likes read, swim, and dance.

Explanation: In a list, all items joined by 'and' should share the same grammatical form; using the gerund form ('-ing') for all three items keeps the list parallel.

Q56. Choose the correctly punctuated sentence for the following direct speech.

HARD

- A. Riya said, I am going home.
- B. Riya said, "I am going home." (Correct)**
- C. Riya said "I am going home".
- D. "Riya said, I am going home."

Explanation: Direct speech needs a comma after the reporting verb, quotation marks around the exact words spoken, and the closing period inside the final quotation mark.

Q57. Choose the correct reported (indirect) speech for: She said, "I am tired."

HARD

- A. She said that she is tired.
- B. She said that she was tired. (Correct)**
- C. She said that she were tired.
- D. She said that she has tired.

Explanation: When the reporting verb 'said' is in the past tense, the present tense in the direct speech ('am') shifts back to the past tense ('was') in reported speech.

Q58. Choose the correct passive voice form of the sentence: 'They had completed the project before the deadline.'

HARD

- A. The project has been completed before the deadline.
- B. The project had been completed before the deadline. (Correct)**
- C. The project had been completing before the deadline.
- D. The project was completed before the deadline.

Explanation: The original sentence is in the past perfect tense, describing an action completed before another past point (the deadline). The passive form must preserve this: 'had been + past participle' gives 'had been completed'.

Q59. Read the passage and answer the question:

"Memory, Nikhil often thought, was a dishonest archivist, filing away only the fragments that suited its mood. The house he had grown up in existed now only in scattered, golden fragments — a creaking stair, the smell of rain on the courtyard tiles — while entire rooms of his childhood had simply vanished from recall. He wondered whether this selective forgetting was mercy or theft. Either way, he had stopped trying to reclaim what memory had quietly buried."

What does the passage suggest about Nikhil's view of memory?

HARD

- A. He believes memory records every event with perfect accuracy.
- B. He has completely forgotten his childhood home.
- C. He sees memory as selective, preserving only certain fragments of the past. (Correct)**
- D. He is actively trying to recover every lost memory.

Explanation: The passage calls memory 'a dishonest archivist, filing away only the fragments that suited its mood' and describes 'selective forgetting' — directly stating that memory preserves only certain fragments, not everything.

Q60. Convert to indirect speech: He said, 'I can help you tomorrow.'

HARD

- A. He said that he can help me tomorrow.
- B. He said that he could help me tomorrow.
- C. He said that he can help me the next day.
- D. He said that he could help me the next day. (Correct)**

Explanation: Since 'said' is in the past tense, the modal 'can' backshifts to 'could', and the time expression 'tomorrow' changes to 'the next day' because the speech is now being reported later — both changes appear only in this option.

Q61. Read the passage and answer the question:

"Memory, Nikhil often thought, was a dishonest archivist, filing away only the fragments that suited its mood. The house he had grown up in existed now only in scattered, golden fragments — a creaking stair, the smell of rain on the courtyard tiles — while entire rooms of his childhood had simply vanished from recall. He wondered whether this selective forgetting was mercy or theft. Either way, he had stopped trying to reclaim what memory had quietly buried."

The tone of the passage can best be described as:

HARD

- A. thoughtful and contemplative (Correct)**
- B. comic and mocking
- C. urgent and alarmed
- D. dismissive and indifferent

Explanation: The passage shows Nikhil quietly reflecting on the nature of memory, wondering 'whether this selective forgetting was mercy or theft' — this unhurried, questioning quality is thoughtful and contemplative.

Q62. Identify the type of clause: 'Although it was raining, they continued to play.'

HARD

- A. Noun clause
- B. Adverb clause of concession (Correct)**
- C. Adjective clause
- D. Adverb clause of time

Explanation: 'Although' introduces a clause showing an unexpected contrast between the rain and their continuing to play — this contrast function defines an adverb clause of concession.

Q63. Read the passage and answer the question:

"Memory, Nikhil often thought, was a dishonest archivist, filing away only the fragments that suited its mood. The house he had grown up in existed now only in scattered, golden fragments — a creaking stair, the smell of rain on the courtyard tiles — while entire rooms of his childhood had simply vanished from recall. He wondered whether this selective forgetting was mercy or theft. Either way, he had stopped trying to reclaim what memory had quietly buried."

Calling memory 'a dishonest archivist' is an example of:

HARD

- A. simile, because it uses 'like' or 'as'
- B. onomatopoeia, because it imitates a sound
- C. metaphor, because memory is directly described as being an archivist (Correct)**
- D. hyperbole, because it is a literal exaggeration of size

Explanation: A metaphor directly states that one thing is another without using 'like' or 'as'. Memory is directly called 'a dishonest archivist', making this a metaphor, not a comparison using connecting words.

Q64. Choose the correct verb to complete the sentence: 'Neither the captain nor the players ____ satisfied with the result.'

HARD

- A. was
- B. is
- C. has been
- D. were (Correct)**

Explanation: With correlative conjunctions like 'neither...nor', the verb agrees with the subject nearest to it. Here, the nearer subject is the plural 'players', so the plural verb 'were' is correct.

General Knowledge (41 questions)

Q1. Which animal is recognised as the National Animal of India?

EASY

- A. Lion
- B. Tiger (Correct)**
- C. Elephant
- D. Leopard

Explanation: The Bengal Tiger is India's National Animal, chosen for its strength and grace, and India runs 'Project Tiger' for its conservation.

Q2. What is the capital city of India?

EASY

- A. Mumbai
- B. New Delhi (Correct)**
- C. Kolkata
- D. Chennai

Explanation: New Delhi is the capital of India, seat of the Parliament and the central government.

Q3. Which is the largest planet in our Solar System?

EASY

- A. Saturn
- B. Earth
- C. Jupiter (Correct)**
- D. Neptune

Explanation: Jupiter is the largest planet in the Solar System by both size and mass.

Q4. Which bird is the National Bird of India?

EASY

- A. Parrot
- B. Peacock (Correct)**
- C. Sparrow
- D. Crow

Explanation: The Indian Peacock is the National Bird of India, chosen for its vibrant colours and cultural significance.

Q5. How many continents are there on Earth?

EASY

- A. 5
- B. 6
- C. 7 (Correct)**
- D. 8

Explanation: Earth is divided into seven continents: Asia, Africa, North America, South America, Antarctica, Europe, and Australia.

Q6. From which direction does the Sun rise every day?

EASY

- A. North
- B. South
- C. East (Correct)**
- D. West

Explanation: Due to Earth's rotation, the Sun appears to rise in the east and set in the west.

Q7. Which sport is popularly known as the 'Gentleman's Game'?

EASY

- A. Football
- B. Hockey
- C. Cricket (Correct)**
- D. Tennis

Explanation: Cricket has traditionally been nicknamed the 'Gentleman's Game' because of its emphasis on fair play and sportsmanship.

Q8. What is the capital city of France?

EASY

- A. London
- B. Paris (Correct)**
- C. Rome
- D. Berlin

Explanation: Paris is the capital city of France, home to landmarks like the Eiffel Tower.

Q9. Which flower is the National Flower of India?

EASY

- A. Rose
- B. Sunflower
- C. Lotus (Correct)**
- D. Marigold

Explanation: The Lotus is India's National Flower, symbolising purity and prosperity in Indian culture.

Q10. What is the largest organ of the human body?

EASY

- A. Liver
- B. Heart
- C. Skin (Correct)**
- D. Brain

Explanation: The skin is the largest organ of the human body, covering and protecting the entire body.

Q11. Which is the largest planet in our solar system?

EASY

- A. Earth
- B. Jupiter (Correct)**
- C. Mars
- D. Venus

Explanation: Jupiter is the largest planet in the solar system by both mass and volume.

Q12. Which river has been declared the National River of India?

MEDIUM

- A. Yamuna
- B. Godavari
- C. Ganga (Correct)**
- D. Brahmaputra

Explanation: The Ganga (Ganges) was declared the National River of India in 2008 due to its cultural, historical, and ecological importance.

Q13. Who founded the Scout movement in 1907?

MEDIUM

- A. Henry Dunant
- B. Robert Baden-Powell (Correct)**
- C. Pierre de Coubertin
- D. Alfred Nobel

Explanation: Robert Baden-Powell, a British Army officer, founded the Scout movement in 1907 to build character and outdoor skills in young people.

Q14. How many players are there in a cricket team on the field during a match?

MEDIUM

- A. 9
- B. 10
- C. 11 (Correct)**
- D. 12

Explanation: A cricket team fields 11 players at a time during a match.

Q15. Which is traditionally regarded as the longest river in the world?

MEDIUM

- A. Amazon
- B. Nile (Correct)**
- C. Yangtze
- D. Mississippi

Explanation: The Nile River in Africa is traditionally taught as the longest river in the world.

Q16. What is the capital city of Japan?

MEDIUM

- A. Osaka
- B. Tokyo (Correct)**
- C. Kyoto
- D. Seoul

Explanation: Tokyo is the capital city of Japan.

Q17. Which natural source is the body's main way of producing Vitamin D?

MEDIUM

- A. Sunlight (Correct)**
- B. Drinking water
- C. Table salt
- D. Moonlight

Explanation: When skin is exposed to sunlight, the body naturally produces Vitamin D.

Q18. Who founded the International Red Cross movement?

MEDIUM

- A. Henry Dunant (Correct)**
- B. Florence Nightingale
- C. Louis Pasteur
- D. Alexander Fleming

Explanation: Henry Dunant, a Swiss businessman, founded the International Red Cross in 1863 after witnessing the suffering of wounded soldiers.

Q19. Which is the smallest continent by land area?

MEDIUM

- A. Europe
- B. Antarctica
- C. Australia (Correct)**
- D. South America

Explanation: Australia is the smallest of the seven continents by land area.

Q20. How many players does one team have on the field in a standard game of football (soccer)?

MEDIUM

- A. 9
- B. 10
- C. 11 (Correct)**
- D. 12

Explanation: A standard football (soccer) team fields 11 players, including the goalkeeper.

Q21. In which Indian city is the Taj Mahal located?

MEDIUM

- A. Delhi
- B. Agra (Correct)**
- C. Jaipur
- D. Lucknow

Explanation: The Taj Mahal is located in Agra, Uttar Pradesh, built by Mughal emperor Shah Jahan.

Q22. Who was the first Prime Minister of independent India?

MEDIUM

- A. Mahatma Gandhi
- B. Jawaharlal Nehru (Correct)**
- C. Sardar Vallabhbhai Patel
- D. Dr. Rajendra Prasad

Explanation: Jawaharlal Nehru became India's first Prime Minister after independence in 1947.

Q23. Which is the largest hot desert in the world?

MEDIUM

- A. Thar Desert
- B. Sahara Desert (Correct)**
- C. Gobi Desert
- D. Arabian Desert

Explanation: The Sahara Desert in Africa is the largest hot desert in the world.

Q24. Which gas do green plants absorb from the air to make their food during photosynthesis?

MEDIUM

- A. Oxygen
- B. Carbon dioxide (Correct)**
- C. Nitrogen
- D. Hydrogen

Explanation: Green plants absorb carbon dioxide from the air and use sunlight to convert it into food, releasing oxygen as a by-product.

Q25. After how many years are the Summer Olympic Games usually held?

MEDIUM

- A. 2 years
- B. 3 years
- C. 4 years (Correct)**
- D. 5 years

Explanation: The Summer Olympic Games are traditionally held every four years.

Q26. What is the capital city of Australia?

MEDIUM

- A. Sydney
- B. Melbourne
- C. Canberra (Correct)**
- D. Perth

Explanation: Canberra is the capital city of Australia, specifically built to serve as the nation's capital.

Q27. Which country has the longest written constitution in the world?

HARD

- A. United States
- B. India (Correct)**
- C. United Kingdom
- D. Canada

Explanation: India's Constitution is the longest written constitution of any sovereign country in the world, containing a large number of articles and schedules.

Q28. How many Fundamental Rights are currently guaranteed by the Indian Constitution?

HARD

- A. 5
- B. 6 (Correct)**
- C. 7
- D. 8

Explanation: The Indian Constitution currently guarantees 6 Fundamental Rights, after the Right to Property was removed from the list by the 44th Constitutional Amendment in 1978.

Q29. Who was the first President of India?

HARD

- A. Dr. Rajendra Prasad (Correct)**
- B. Dr. S. Radhakrishnan
- C. Zakir Husain
- D. V.V. Giri

Explanation: Dr. Rajendra Prasad served as the first President of India from 1950 to 1962.

Q30. Which is the tallest mountain peak in the world above sea level?

HARD

- A. K2
- B. Kangchenjunga
- C. Mount Everest (Correct)**
- D. Nanga Parbat

Explanation: Mount Everest, located on the border of Nepal and Tibet, is the tallest mountain peak above sea level.

Q31. In India's three-tier Panchayati Raj system, what is the middle (intermediate) level called?

HARD

- A. Gram Sabha
- B. Panchayat Samiti (Correct)**
- C. Zila Parishad
- D. Lok Sabha

Explanation: The three-tier Panchayati Raj system consists of the Gram Panchayat (village level), Panchayat Samiti (block/intermediate level), and Zila Parishad (district level); Panchayat Samiti is the middle tier.

Q32. Who discovered the ABO human blood group system?

HARD

- A. Karl Landsteiner (Correct)**
- B. Louis Pasteur
- C. Edward Jenner
- D. Gregor Mendel

Explanation: Austrian scientist Karl Landsteiner discovered the ABO blood group system in 1901, a finding for which he later won the Nobel Prize.

Q33. How many bones are there in an average adult human body?

HARD

- A. 186
- B. 206 (Correct)**
- C. 226
- D. 246

Explanation: An average adult human body has 206 bones (babies are born with more bones that fuse together as they grow).

Q34. What is the approximate speed of light in vacuum?

HARD

- A. 3 lakh km per second (Correct)**
- B. 3 thousand km per second
- C. 30 lakh km per second
- D. 3 lakh km per hour

Explanation: Light travels at approximately 3,00,000 kilometres (3 lakh km) per second in a vacuum.

Q35. Who was the first human to walk on the Moon?

HARD

- A. Yuri Gagarin
- B. Neil Armstrong (Correct)**
- C. Buzz Aldrin
- D. Michael Collins

Explanation: Neil Armstrong became the first human to walk on the Moon on 20 July 1969 during the Apollo 11 mission.

Q36. N.R. Narayana Murthy is best known as a co-founder of which major Indian IT company?

HARD

- A. Wipro
- B. Infosys (Correct)**
- C. TCS
- D. HCL

Explanation: N.R. Narayana Murthy is one of the co-founders of Infosys, established in 1981, one of India's leading IT services companies.

Q37. Who wrote India's National Anthem, 'Jana Gana Mana'?

HARD

- A. Bankim Chandra Chattopadhyay
- B. Rabindranath Tagore (Correct)**
- C. Sarojini Naidu
- D. Muhammad Iqbal

Explanation: Rabindranath Tagore wrote and composed 'Jana Gana Mana', which was adopted as India's National Anthem.

Q38. How many spokes does the Ashoka Chakra on the Indian national flag have?

HARD

- A. 20
- B. 22
- C. 24 (Correct)**
- D. 26

Explanation: The Ashoka Chakra at the centre of the Indian national flag has 24 spokes.

Q39. Which Indian state has the longest coastline?

HARD

- A. Tamil Nadu
- B. Gujarat (Correct)**
- C. Andhra Pradesh
- D. Maharashtra

Explanation: Gujarat has the longest coastline among Indian states, stretching along the Arabian Sea.

Q40. Which planet is popularly known as the 'Red Planet'?

HARD

- A. Venus
- B. Mars (Correct)**
- C. Mercury
- D. Saturn

Explanation: Mars is called the 'Red Planet' because of the reddish appearance caused by iron oxide (rust) on its surface.

Q41. What is the chemical symbol for the element Gold?

HARD

- A. Go
- B. Gd
- C. Au (Correct)**
- D. Ag

Explanation: The chemical symbol for Gold is 'Au', derived from its Latin name 'aurum'.

Mathematics (73 questions)

Q1. Solve for x: $2x + 3 = 11$.

EASY

- A. 5.5
- B. 4 (Correct)**
- C. 7
- D. 8

Explanation: $2x + 3 = 11$ ' $2x = 11 - 3 = 8$ ' $x = 8 \div 2 = 4$.

Q2. Simplify: $3(x + 4) - 2x$.

EASY

- A. $5x + 12$
- B. $x + 36$
- C. $x + 12$ (Correct)**
- D. $x + 4$

Explanation: $3(x+4) = 3x + 12$. Then $3x + 12 - 2x = x + 12$.

Q3. Find 20% of 150.

EASY

- A. 15
- B. 20
- C. 3
- D. 30 (Correct)**

Explanation: $20\% \text{ of } 150 = (20/100) \times 150 = 30$.

Q4. Express the ratio 12 : 18 in its simplest form.

EASY

- A. 2 : 3 (Correct)**
- B. 3 : 2
- C. 6 : 9
- D. 4 : 6

Explanation: HCF of 12 and 18 is 6. Dividing both terms by 6: $12 \div 6 : 18 \div 6 = 2 : 3$.

Q5. Find the simple interest on Rs 1000 for 2 years at 5% per annum.

EASY

- A. 10000
- B. 100 (Correct)**
- C. 50
- D. 1100

Explanation: $SI = (P \times R \times T)/100 = (1000 \times 5 \times 2)/100 = 10000/100 = 100$.

Q6. Find the perimeter of a rectangle with length 12 cm and breadth 5 cm.

EASY

- A. 60
- B. 68
- C. 34 (Correct)**
- D. 17

Explanation: Perimeter = $2(l + b) = 2(12 + 5) = 2 \times 17 = 34$ cm.

Q7. Find the area of a square with side 7 cm.

EASY

- A. 28
- B. 14
- C. 63
- D. 49 (Correct)**

Explanation: Area of square = side $\times$ side = $7 \times 7 = 49$ cm².

Q8. In a triangle, two of the angles are 50° and 60°. Find the third angle.

EASY

- A. 70 (Correct)**
- B. 110
- C. 130
- D. 120

Explanation: Sum of angles of a triangle = 180°. Third angle = $180 - 50 - 60 = 70^\circ$.

Q9. Rohan buys 5 pens at Rs 8 each and 3 notebooks at Rs 15 each. Find the total amount he pays.

EASY

- A. 40
- B. 85 (Correct)**
- C. 99
- D. 45

Explanation: Cost of pens = $5 \times 8 = 40$. Cost of notebooks = $3 \times 15 = 45$. Total = $40 + 45 = 85$.

Q10. If 4 books cost Rs 200, find the cost of 7 books at the same rate.

EASY

- A. 400
- B. 114.29
- C. 350 (Correct)**
- D. 300

Explanation: Cost of 1 book = $200 \div 4 = 50$. Cost of 7 books = $50 \times 7 = 350$.

Q11. Solve for x: $2x + 3 = 11$

EASY

- A. 4 (Correct)**
- B. 8
- C. 7
- D. 6

Explanation: $2x = 11 - (3) = 8$, so $x = 8 \div 2 = 4$.

Q12. Solve for x: $3x - 4 = 11$

EASY

- A. 15
- B. 5 (Correct)**
- C. 2.3333333333333335
- D. 8

Explanation: $3x = 11 - (-4) = 15$, so $x = 15 \div 3 = 5$.

Q13. Solve for x: $4x + 5 = 25$

EASY

- A. 20
- B. 7.5
- C. 5 (Correct)**
- D. 9

Explanation: $4x = 25 - (5) = 20$, so $x = 20 \div 4 = 5$.

Q14. Solve for x: $5x - 6 = 24$

EASY

- A. 30
- B. 3.6
- C. 11
- D. 6 (Correct)**

Explanation: $5x = 24 - (-6) = 30$, so $x = 30 \div 5 = 6$.

Q15. Solve for x: $6x + 7 = 43$

EASY

- A. 6 (Correct)**
- B. 36
- C. 8.333333333333334
- D. 12

Explanation: $6x = 43 - (7) = 36$, so $x = 36 \div 6 = 6$.

Q16. Solve for x: $2x - 9 = 5$

EASY

- A. 14
- B. 7 (Correct)**
- C. -2
- D. 9

Explanation: $2x = 5 - (-9) = 14$, so $x = 14 \div 2 = 7$.

Q17. Solve for x: $3x + 8 = 26$

EASY

- A. 18
- B. 11.333333333333334
- C. 6 (Correct)**
- D. 9

Explanation: $3x = 26 - (8) = 18$, so $x = 18 \div 3 = 6$.

Q18. Solve for x: $7x - 2 = 33$

EASY

- A. 35
- B. 4.428571428571429
- C. 12
- D. 5 (Correct)**

Explanation: $7x = 33 - (-2) = 35$, so $x = 35 \div 7 = 5$.

Q19. Solve for x: $4x - 3 = 17$

EASY

- A. 5 (Correct)**
- B. 20
- C. 3.5
- D. 9

Explanation: $4x = 17 - (-3) = 20$, so $x = 20 \div 4 = 5$.

Q20. Solve for x: $5x + 10 = 45$

EASY

- A. 35
- B. 7 (Correct)**
- C. 11
- D. 12

Explanation: $5x = 45 - (10) = 35$, so $x = 35 \div 5 = 7$.

Q21. Evaluate: $\frac{1}{2} + \frac{1}{3}$

EASY

- A. $\frac{5}{6}$ (Correct)**
- B. $\frac{1}{6}$
- C. $\frac{2}{5}$
- D. 1

Explanation: Converting to a common denominator of 6: the sum simplifies to $\frac{5}{6}$.

Q22. Evaluate: $-\frac{2}{3} + \frac{1}{4}$

EASY

- A. $-\frac{11}{12}$
- B. $-\frac{5}{12}$ (Correct)**
- C. $-\frac{1}{7}$
- D. $-\frac{1}{3}$

Explanation: Converting to a common denominator of 12: the sum simplifies to $-\frac{5}{12}$.

Q23. Evaluate: $\frac{3}{4} - \frac{1}{6}$

EASY

- A. $\frac{11}{12}$
- B. $\frac{2}{5}$
- C. $\frac{7}{12}$ (Correct)**
- D. $\frac{1}{2}$

Explanation: Converting to a common denominator of 24: the sum simplifies to $\frac{7}{12}$.

Q24. Evaluate: $-\frac{1}{2} - \frac{1}{3}$

EASY

- A. $-\frac{1}{6}$
- B. 0
- C. -1
- D. $-\frac{5}{6}$ (Correct)**

Explanation: Converting to a common denominator of 6: the sum simplifies to $-\frac{5}{6}$.

Q25. Evaluate: $\frac{2}{5} + \frac{3}{10}$

EASY

- A. $\frac{7}{10}$ (Correct)**
- B. $\frac{1}{10}$
- C. $\frac{1}{3}$
- D. 1

Explanation: Converting to a common denominator of 10: the sum simplifies to $\frac{7}{10}$.

Q26. Evaluate: $-\frac{3}{8} + \frac{1}{2}$

EASY

- A. $-\frac{7}{8}$
- B. $\frac{1}{8}$ (Correct)**
- C. $-\frac{1}{5}$
- D. $-\frac{1}{4}$

Explanation: Converting to a common denominator of 8: the sum simplifies to $\frac{1}{8}$.

Q27. Evaluate: $\frac{5}{6} - \frac{1}{2}$

EASY

- A. $\frac{4}{3}$
- B. $\frac{3}{4}$
- C. $\frac{1}{3}$ (Correct)**
- D. $\frac{2}{3}$

Explanation: Converting to a common denominator of 6: the sum simplifies to $\frac{1}{3}$.

Q28. Evaluate: $-\frac{1}{4} - \frac{1}{8}$

EASY

- A. $-\frac{1}{8}$
- B. 0
- C. $-\frac{1}{2}$
- D. $-\frac{3}{8}$ (Correct)**

Explanation: Converting to a common denominator of 8: the sum simplifies to $-\frac{3}{8}$.

Q29. Evaluate: $\frac{7}{9} - \frac{2}{3}$

EASY

- A. $\frac{1}{9}$ (Correct)**
- B. $\frac{13}{9}$
- C. $\frac{3}{4}$
- D. $\frac{5}{9}$

Explanation: Converting to a common denominator of 9: the sum simplifies to $\frac{1}{9}$.

Q30. Solve for x: $3x - 5 = 2x + 7$.

MEDIUM

- A. 2
- B. -12
- C. 6
- D. 12 (Correct)**

Explanation: $3x - 2x = 7 + 5$ 'x' = 12.

Q31. Simplify: $5x - 3(2x - 4)$.

MEDIUM

- A. $-x + 12$ (Correct)**
- B. $-x - 12$
- C. $-x$
- D. $11x - 12$

Explanation: $5x - 3(2x-4) = 5x - 6x + 12 = -x + 12$.

Q32. A number 250 is increased by 12%. Find the new number.

MEDIUM

- A. 220
- B. 280 (Correct)**
- C. 262
- D. 275

Explanation: $12\% \text{ of } 250 = 30$. New number $= 250 + 30 = 280$.

Q33. 25% of a certain number is 40. Find the number.

MEDIUM

- A. 65
- B. 50
- C. 160 (Correct)**
- D. 10

Explanation: Let the number be x . $0.25x = 40$ ' $x = 40 \div 0.25 = 160$.

Q34. Rs 360 is divided among three people in the ratio 2 : 3 : 4. Find the largest share.

MEDIUM

- A. 120
- B. 40
- C. 180
- D. 160 (Correct)**

Explanation: Sum of ratio terms $= 2+3+4 = 9$. Value of one part $= 360/9 = 40$. Largest share $= 4 \times 40 = 160$.

Q35. A sum of Rs 1500 earns simple interest of Rs 225 in 3 years. Find the rate of interest per annum.

MEDIUM

- A. 5 (Correct)**
- B. 15
- C. 0.05
- D. 7.5

Explanation: $R = (SI \times 100)/(P \times T) = (225 \times 100)/(1500 \times 3) = 22500/4500 = 5\%$.

Q36. Find the compound interest on Rs 2000 for 2 years at 10% per annum, compounded annually.

MEDIUM

- A. 662
- B. 420 (Correct)**
- C. 400
- D. 2420

Explanation: Amount $= 2000 \times (1.1)^2 = 2000 \times 1.21 = 2420$. CI = Amount - Principal $= 2420 - 2000 = 420$.

Q37. Find the area of a parallelogram with base 10 cm and height 6 cm.

MEDIUM

- A. 30
- B. 16
- C. 60 (Correct)**
- D. 32

Explanation: Area of parallelogram $= \text{base} \times \text{height} = 10 \times 6 = 60 \text{ cm}^2$.

Q38. Find the area of a circle of radius 7 cm. (Use $\pi = 22/7$.)

MEDIUM

- A. 44
- B. 22
- C. 616
- D. 154 (Correct)**

Explanation: Area $= \pi r^2 = (22/7) \times 7 \times 7 = 22 \times 7 = 154 \text{ cm}^2$.

Q39. The perimeter of a rectangle is 54 cm and its length is 17 cm. Find its breadth.

MEDIUM

- A. 10 (Correct)**
- B. 37
- C. 44
- D. 13.5

Explanation: $\text{Perimeter} = 2(l+b)$ ' $54/2 = 27 = l + b$ ' $b = 27 - 17 = 10 \text{ cm}$.

Q40. Two angles are complementary. If one angle is 34° , find the other angle.

MEDIUM

- A. 66
- B. 56 (Correct)**
- C. 146
- D. 34

Explanation: Complementary angles add up to 90° . Other angle = $90 - 34 = 56^\circ$.

Q41. A father's age is 3 times his son's age. The sum of their ages is 48 years. Find the son's age.

MEDIUM

- A. 36
- B. 24
- C. 12 (Correct)**
- D. 16

Explanation: Let son's age = x , father's age = $3x$. $x + 3x = 48$ ' $4x = 48$ ' $x = 12$.

Q42. Find the average of the numbers 10, 20, 25, 35 and 40.

MEDIUM

- A. 25
- B. 40
- C. 32.5
- D. 26 (Correct)**

Explanation: $\text{Sum} = 10+20+25+35+40 = 130$. $\text{Average} = 130 \div 5 = 26$.

Q43. Find the volume of a cuboid with length 8 cm, breadth 5 cm and height 3 cm.

MEDIUM

- A. 120 (Correct)**
- B. 40
- C. 16
- D. 158

Explanation: $\text{Volume} = l \times b \times h = 8 \times 5 \times 3 = 120 \text{ cm}^3$.

Q44. A car travels 180 km in 3 hours at a constant speed. How far will it travel in 5 hours at the same speed?

MEDIUM

- A. 360
- B. 300 (Correct)**
- C. 108
- D. 240

Explanation: $\text{Speed} = 180/3 = 60 \text{ km/h}$. $\text{Distance in 5 hours} = 60 \times 5 = 300 \text{ km}$.

Q45. Simplify: $3(x + 4) - 2x = ?$

MEDIUM

- A. $x + 12$ (Correct)**
- B. $x - 12$
- C. $5x + 12$
- D. $x + 4$

Explanation: Distribute: $3x + 12 - 2x = x + 12$.

Q46. The value of 225 is:

MEDIUM

- A. 12
- B. 15 (Correct)**
- C. 18
- D. 25

Explanation: $15 \times 15 = 225$, so $225 = 15$.

Q47. If $p(x) = 1x^2 + 2x + 3$, find $p(4)$.

MEDIUM

- A. 27 (Correct)**
- B. 15
- C. 11
- D. 24

Explanation: $p(4) = 1 \times 4^2 + 2 \times 4 + 3 = 16 + 8 + 3 = 27$.

Q48. If $p(x) = 2x^2 - 3x + 5$, find $p(3)$.

MEDIUM

- A. 2
- B. 14 (Correct)**
- C. 32
- D. 9

Explanation: $p(3) = 2 \times 3^2 - 3 \times 3 + 5 = 18 - 9 + 5 = 14$.

Q49. If $p(x) = 1x^2 - 4x + 6$, find $p(5)$.

MEDIUM

- A. -9
- B. 51
- C. 11 (Correct)**
- D. 5

Explanation: $p(5) = 1 \times 5^2 - 4 \times 5 + 6 = 25 - 20 + 6 = 11$.

Q50. If $p(x) = 3x^2 + 2x - 1$, find $p(2)$.

MEDIUM

- A. 9
- B. 7
- C. 16
- D. 15 (Correct)**

Explanation: $p(2) = 3 \times 2^2 + 2 \times 2 - 1 = 12 + 4 - 1 = 15$.

Q51. If $p(x) = 2x^2 + 5x + 1$, find $p(3)$.

MEDIUM

- A. 34 (Correct)**
- B. 22
- C. 4
- D. 33

Explanation: $p(3) = 2 \times 3^2 + 5 \times 3 + 1 = 18 + 15 + 1 = 34$.

Q52. Solve for x: $x/2 + x/3 = 10$.

HARD

- A. 60
- B. 2
- C. 12 (Correct)**
- D. 25

Explanation: LCM of 2 and 3 is 6: $3x/6 + 2x/6 = 10 \Rightarrow 5x/6 = 10 \Rightarrow x = 60/5 = 12$.

Q53. Simplify: $2(3x - 4) - 3(x - 2) + 5$.

HARD

- A. $3x - 9$
- B. $3x - 2$
- C. $3x + 7$
- D. $3x + 3$ (Correct)**

Explanation: $2(3x-4) = 6x - 8$. $-3(x-2) = -3x + 6$. Sum: $6x - 8 - 3x + 6 + 5 = 3x + 3$.

Q54. A number 500 is first increased by 20% and then the result is decreased by 10%. Find the final number.

HARD

- A. 540 (Correct)**
- B. 550
- C. 660
- D. 450

Explanation: After 20% increase: $500 \times 1.20 = 600$. After 10% decrease: $600 \times 0.90 = 540$.

Q55. The cost price of an article is Rs 800 and it is sold for Rs 920. Find the profit percentage.

HARD

- A. 115
- B. 15 (Correct)**
- C. 13.04
- D. 86.96

Explanation: Profit = $920 - 800 = 120$. Profit % = $(120/800) \times 100 = 15\%$.

Q56. Three numbers are in the ratio 3 : 4 : 5. The sum of the largest and the smallest number is 40. Find the middle number.

HARD

- A. 15
- B. 13.33
- C. 20 (Correct)**
- D. 25

Explanation: Let the numbers be $3x, 4x, 5x$. Largest + smallest = $5x + 3x = 8x = 40$ 'x = 5. Middle number = $4x = 20$.

Q57. The compound interest on a certain sum for 2 years at 10% per annum, compounded annually, is Rs 210. Find the sum.

HARD

- A. 2100
- B. 1050
- C. 2000
- D. 1000 (Correct)**

Explanation: $CI = P[(1.1)^2 - 1] = P \times 0.21 = 210$ 'P = $210 \div 0.21 = 1000$.

Q58. Find the difference between the compound interest and the simple interest on Rs 5000 for 2 years at 8% per annum.

HARD

- A. 32 (Correct)**
- B. 800
- C. 832
- D. 64

Explanation: $SI = (5000 \times 8 \times 2)/100 = 800$. $CI = 5000 \times [(1.08)^2 - 1] = 5000 \times 0.1664 = 832$. Difference = $832 - 800 = 32$.

Q59. Find the area of a triangle whose sides are 13 cm, 14 cm and 15 cm, using Heron's formula.

HARD

- A. 21
- B. 84 (Correct)**
- C. 91
- D. 7056

Explanation: $s = (13+14+15)/2 = 21$. Area = $[s(s-a)(s-b)(s-c)] = [21 \times 8 \times 7 \times 6] = 7056 = 84 \text{ cm}^2$.

Q60. The area of a circle is 154 cm^2 . Find its radius. (Use $\pi = 22/7$.)

HARD

- A. 24.5
- B. 14
- C. 7 (Correct)**
- D. 49

Explanation: $\pi r^2 = 154$ ' $r^2 = 154 \times 7/22 = 49$ ' $r = 49 = 7 \text{ cm}$.

Q61. The total surface area of a cube is 96 cm^2 . Find its volume.

HARD

- A. 16
- B. 4
- C. 384
- D. 64 (Correct)**

Explanation: $6a^2 = 96$ ' $a^2 = 16$ ' $a = 4 \text{ cm}$. Volume = $a^3 = 4^3 = 64 \text{ cm}^3$.

Q62. The cost of 3 pens and 2 pencils is Rs 39, and the cost of 5 pens and 3 pencils is Rs 62. Find the cost of one pen.

HARD

- A. 7 (Correct)**
- B. 9
- C. 8
- D. 11.5

Explanation: $3p+2c=39$...(1); $5p+3c=62$...(2). Multiply (1) by 3 and (2) by 2: $9p+6c=117$ and $10p+6c=124$. Subtracting: $p = 7$.

Q63. A can complete a piece of work in 10 days and B can complete it in 15 days. Working together, in how many days will they complete the work?

HARD

- A. 5
- B. 6 (Correct)**
- C. 25
- D. 12.5

Explanation: A's rate = $1/10$, B's rate = $1/15$. Combined rate = $1/10 + 1/15 = 3/30 + 2/30 = 5/30 = 1/6$. Time = 6 days.

Q64. A car travels 60 km at a speed of 30 km/h and the next 60 km at a speed of 60 km/h. Find its average speed for the whole journey.

HARD

- A. 30
- B. 20
- C. 40 (Correct)**
- D. 45

Explanation: Time for first 60 km = $60/30 = 2 \text{ h}$. Time for next 60 km = $60/60 = 1 \text{ h}$. Total distance = 120 km, total time = 3 h. Average speed = $120/3 = 40 \text{ km/h}$.

Q65. A and B start a business investing Rs 5000 and Rs 7000 respectively. If the profit at the end of the year is Rs 3600, divided in the ratio of their investments, find A's share.

HARD

- A. 2100
- B. 1800
- C. 1200
- D. 1500 (Correct)**

Explanation: Ratio of investments = 5000:7000 = 5:7, total parts = 12. A's share = $3600 \times 5/12 = 1500$.

Q66. An article has a marked price of Rs 2000. A discount of 10% is given, and then 5% GST is charged on the discounted price. Find the final price paid.

HARD

- A. 1890 (Correct)**
- B. 1900
- C. 1700
- D. 2100

Explanation: Discounted price = $2000 \times 0.90 = 1800$. GST = $1800 \times 0.05 = 90$. Final price = $1800 + 90 = 1890$.

Q67. Using the identity $(ab)(a+b) = a^2b^2$, quickly find 47×53 .

HARD

- A. 2491 (Correct)**
- B. 2509
- C. 2497
- D. 2482

Explanation: $47 = 50 - 3$ and $53 = 50 + 3$, so $47 \times 53 = 50^2 - 3^2 = 2500 - 9 = 2491$.

Q68. Using the identity $(ab)(a+b) = a^2b^2$, quickly find 96×104 .

HARD

- A. 10016
- B. 9984 (Correct)**
- C. 9996
- D. 9968

Explanation: $96 = 100 - 4$ and $104 = 100 + 4$, so $96 \times 104 = 100^2 - 4^2 = 10000 - 16 = 9984$.

Q69. Using the identity $(ab)(a+b) = a^2b^2$, quickly find 38×42 .

HARD

- A. 1604
- B. 1598
- C. 1596 (Correct)**
- D. 1592

Explanation: $38 = 40 - 2$ and $42 = 40 + 2$, so $38 \times 42 = 40^2 - 2^2 = 1600 - 4 = 1596$.

Q70. Using the identity $(ab)(a+b) = a^2b^2$, quickly find 55×65 .

HARD

- A. 3625
- B. 3595
- C. 3550
- D. 3575 (Correct)**

Explanation: $55 = 60 - 5$ and $65 = 60 + 5$, so $55 \times 65 = 60^2 - 5^2 = 3600 - 25 = 3575$.

Q71. Using the identity $(ab)(a+b) = a^2b^2$, quickly find 194×206 .

HARD

- A. 39964 (Correct)**
- B. 40036
- C. 39994
- D. 39928

Explanation: $194 = 200 - 6$ and $206 = 200 + 6$, so $194 \times 206 = 200^2 - 6^2 = 40000 - 36 = 39964$.

Q72. Using the identity $(ab)(a+b) = a^2b^2$, quickly find 73×87 .

HARD

- A. 6449
- B. 6351 (Correct)**
- C. 6393
- D. 6302

Explanation: $73 = 80 - 7$ and $87 = 80 + 7$, so $73 \times 87 = 80^2 - 7^2 = 6400 - 49 = 6351$.

Q73. Using the identity $(ab)(a+b) = a^2b^2$, quickly find 26×34 .

HARD

- A. 916
- B. 896
- C. 884 (Correct)**
- D. 868

Explanation: $26 = 30 + 4$ and $34 = 30 + 4$, so $26 \times 34 = (30 + 4)^2 = 900 + 16 + 240 = 1156$.

Science (73 questions)

Q1. What is the basic structural and functional unit of all living organisms?

EASY

- A. Cell (Correct)**
- B. Tissue
- C. Organ
- D. Organ system

Explanation: The cell is the smallest unit that can carry out all life processes; all living organisms are made up of one or more cells.

Q2. Green plants prepare their own food through a process that uses sunlight. What is this process called?

EASY

- A. Excretion
- B. Photosynthesis (Correct)**
- C. Respiration
- D. Digestion

Explanation: Photosynthesis is the process by which green plants use sunlight, water, and carbon dioxide to make their own food (glucose) and release oxygen.

Q3. Which organ in the human body is mainly responsible for pumping blood throughout the body?

EASY

- A. Lungs
- B. Stomach
- C. Heart (Correct)**
- D. Liver

Explanation: The heart is a muscular organ that continuously pumps blood through blood vessels to all parts of the body.

Q4. In most plants, which part is mainly responsible for absorbing water and minerals from the soil?

EASY

- A. Leaf
- B. Flower
- C. Stem
- D. Root (Correct)**

Explanation: Roots grow into the soil and absorb water and dissolved minerals, which are then transported upward through the plant.

Q5. Under normal conditions in a single uniform medium (like air), light travels in a

EASY

- A. straight line (Correct)**
- B. circular path
- C. zig-zag path
- D. only downward path

Explanation: Light travels in a straight line in a uniform medium, which explains phenomena such as the formation of shadows.

Q6. A see-saw (a rigid bar that turns around a fixed point to lift a load) is an example of which simple machine?

EASY

- A. Wedge
- B. Lever (Correct)**
- C. Wheel and axle
- D. Screw

Explanation: A lever is a rigid bar that pivots around a fixed point called a fulcrum, used to lift loads with less effort; a see-saw is a classic example.

Q7. Which force pulls objects toward the Earth, causing a dropped ball to fall downward?

EASY

- A. Frictional force
- B. Electric force
- C. Gravitational force (Correct)**
- D. Magnetic force

Explanation: Gravity is the force of attraction that the Earth exerts on objects, pulling them toward its center, which causes unsupported objects to fall downward.

Q8. Ice, water, and water vapor are three different examples of which three states of matter respectively?

EASY

- A. Liquid, gas, solid
- B. Gas, solid, liquid
- C. Solid, gas, liquid
- D. Solid, liquid, gas (Correct)**

Explanation: Ice is the solid state of water, liquid water is the liquid state, and water vapor is the gaseous state — the three common states of matter.

Q9. Which of these common substances tastes sour and is a well-known example of a mild acid?

EASY

- A. Lemon juice (Correct)**
- B. Baking soda solution
- C. Soap solution
- D. Plain water

Explanation: Lemon juice contains citric acid, giving it a sour taste; it is a commonly cited example of a mild natural acid.

Q10. When ice (a solid) is heated and turns into water (a liquid), this process is called

EASY

- A. Condensation
- B. Melting (Correct)**
- C. Freezing
- D. Evaporation

Explanation: Melting is the change of state from solid to liquid that occurs when a solid substance is heated, such as ice turning into water.

Q11. Which gas do plants absorb from the air during photosynthesis?

EASY

- A. Oxygen
- B. Nitrogen
- C. Carbon Dioxide (Correct)**
- D. Hydrogen

Explanation: Plants take in carbon dioxide and release oxygen during photosynthesis.

Q12. What is the SI unit of speed?

EASY

- A. metre per second (m/s) (Correct)**
- B. metre per second squared (m/s^2)
- C. kilogram per cubic metre (kg/m^3)
- D. newton (N)

Explanation: Speed = distance/time, so its SI unit is metre per second (m/s).

Q13. Newton's first law of motion is also known as the law of _____.

EASY

- A. conservation of momentum
- B. inertia (Correct)**
- C. action and reaction
- D. universal gravitation

Explanation: Newton's first law states that a body continues in its state of rest or uniform motion unless acted upon by an external force. This property of a body is called inertia, so the first law is also called the law of inertia.

Q14. Which of the following is the SI unit of weight?

EASY

- A. kilogram
- B. joule
- C. newton (Correct)**
- D. newton-metre

Explanation: Weight is the force of gravity acting on a body ($W = mg$). Since force is measured in newton, weight is also measured in newton.

Q15. The SI unit of work done is the _____.

EASY

- A. watt
- B. newton
- C. pascal
- D. joule (Correct)**

Explanation: Work done = force $\times$ displacement. Since force is in newton and displacement in metre, work is measured in newton-metre, which is given the special name joule.

Q16. Which of the following processes involves a liquid changing directly into vapour without boiling, occurring only at its surface?

EASY

- A. evaporation (Correct)**
- B. boiling
- C. condensation
- D. freezing

Explanation: Evaporation is a surface phenomenon in which liquid particles at the surface gain enough energy to escape into the vapour state at temperatures below the boiling point.

Q17. Which of the following is an example of a homogeneous mixture?

EASY

- A. sand and water
- B. salt solution (Correct)**
- C. oil and water
- D. milk

Explanation: A salt solution is homogeneous because the salt is completely and uniformly dissolved in water, forming a single visible phase throughout.

Q18. What does the chemical formula H_2O tell us about a molecule of water?

EASY

- A. it contains 2 grams of hydrogen and 1 gram of oxygen
- B. it contains 1 atom of hydrogen and 2 atoms of oxygen
- C. it contains 2 atoms of hydrogen and 1 atom of oxygen (Correct)**
- D. it contains 2 molecules of hydrogen and 1 molecule of oxygen

Explanation: The subscript numbers in a chemical formula show the number of atoms of each element present in one molecule; H_2O means 2 hydrogen atoms are bonded to 1 oxygen atom.

Q19. Which cell organelle is known as the 'powerhouse of the cell'?

EASY

- A. nucleus
- B. ribosome
- C. Golgi apparatus
- D. mitochondrion (Correct)**

Explanation: Mitochondria generate ATP through cellular respiration, providing energy for the cell's activities, which is why they are nicknamed the 'powerhouse of the cell.'

Q20. Which plant tissue is responsible for the upward transport of water and minerals from the roots to the leaves?

EASY

- A. xylem (Correct)**
- B. phloem
- C. parenchyma
- D. collenchyma

Explanation: Xylem is a conducting tissue made of vessels and tracheids that transports water and dissolved minerals upward from the roots to the rest of the plant.

Q21. In the five-kingdom classification proposed by Whittaker, which kingdom includes organisms that are unicellular and lack a well-defined nucleus?

EASY

- A. Protista
- B. Monera (Correct)**
- C. Fungi
- D. Plantae

Explanation: Kingdom Monera consists of prokaryotic organisms such as bacteria, whose cells lack a membrane-bound nucleus and other membrane-bound organelles.

Q22. The distance-time graph of an object is a straight line inclined to the time axis. What does this indicate about the object's motion?

EASY

- A. The object is moving with uniform (constant) speed (Correct)**
- B. The object is at rest
- C. The object is moving with non-uniform (changing) speed
- D. The object is moving with uniformly increasing acceleration

Explanation: A straight-line distance-time graph means equal distances are covered in equal intervals of time, i.e., the speed is constant/uniform. A curved graph would instead indicate non-uniform motion.

Q23. Why can we not hear any sound from an explosion happening in outer space (a vacuum), even though it would be extremely loud?

EASY

- A. Because sound waves travel too fast to be detected
- B. Because sound waves are absorbed completely by darkness
- C. Because sound needs a material medium to travel, and a vacuum has no particles to carry the vibrations- (Correct)**
- D. Because the human ear cannot function at all without atmospheric pressure

Explanation: Sound is a mechanical wave that travels by the vibration of particles of a medium (solid, liquid, or gas). A vacuum has no particles at all, so there is nothing to vibrate and carry the sound wave, and hence no sound can travel through it.

Q24. Which of the following is a compound (and not simply an element or a mixture)?

EASY

- A. Carbon dioxide (CO) (Correct)**
- B. Air
- C. Iron (Fe)
- D. Sea water

Explanation: A compound is a pure substance formed of two or more elements chemically combined in a fixed ratio. Carbon dioxide is carbon and oxygen chemically bonded together in a fixed proportion, making it a true compound.

Q25. Which scientist proposed the widely used five-kingdom classification system (Monera, Protista, Fungi, Plantae, Animalia), based on criteria such as cell structure, mode of nutrition, and body organization?

EASY

- A. Carl Linnaeus
- B. Charles Darwin
- C. Gregor Mendel
- D. R. H. Whittaker (Correct)**

Explanation: R. H. Whittaker proposed the five-kingdom classification system in 1969, grouping organisms into Monera, Protista, Fungi, Plantae, and Animalia based on cell structure (prokaryotic or eukaryotic), body organization, mode of nutrition, and other criteria.

Q26. The acceleration due to gravity on the Moon is about one-sixth that on Earth. If a body weighs 60 N on Earth, what will it weigh on the Moon?

EASY

- A. 60 N
- B. 6 N
- C. 360 N
- D. 10 N (Correct)**

Explanation: Weight = mass $\times$ g. Since g on the Moon is about 1/6th of g on Earth, weight on the Moon = $60/6 = 10$ N. The body's mass stays the same; only its weight changes because g is smaller.

Q27. A person feels cold when their wet hands are exposed to air blowing from a fan. What is the main reason for this cooling sensation?

EASY

- A. Evaporation of water from the hands absorbs heat from the body, producing a cooling effect (Correct)**
- B. Moving air itself is always colder in temperature than still air
- C. The fan chemically reacts with the water on the hands to release cold
- D. The water freezes instantly on the skin due to the airflow

Explanation: Evaporation is a cooling process: liquid water molecules absorb energy (latent heat) from the skin in order to change into vapour, which lowers the skin's temperature. Moving air from a fan speeds up evaporation by carrying away vapour molecules from the surface, enhancing the cooling effect.

Q28. A passenger standing in a moving bus falls backward when the bus suddenly starts moving forward from rest. Which concept best explains this?

EASY

- A. Newton's third law of motion
- B. Newton's second law of motion
- C. Conservation of momentum in a colliding system
- D. Newton's first law of motion (inertia of rest) (Correct)**

Explanation: The passenger's body was at rest and, by inertia, tends to remain at rest even as the bus floor moves forward beneath them. Relative to the bus, the passenger therefore appears to fall backward — this is Newton's first law (inertia of rest).

Q29. Diffusion (the natural intermixing of particles of two substances on their own) occurs the fastest in which state of matter?

EASY

- A. Solids
- B. Liquids
- C. Gases (Correct)**
- D. All three states diffuse at exactly the same rate

Explanation: Gas particles have the greatest spacing between them, the highest kinetic energy, and the weakest forces of attraction, so they diffuse the fastest. Liquids diffuse more slowly than gases, and solids show negligible diffusion under ordinary conditions.

Q30. Which of these structures is typically present in a plant cell but absent in an animal cell?

MEDIUM

- A. Cell membrane
- B. Cytoplasm
- C. Cell wall (Correct)**
- D. Nucleus

Explanation: Plant cells have a rigid cell wall (made mainly of cellulose) surrounding the cell membrane, which provides structural support; animal cells lack this cell wall.

Q31. Tiny pores mostly found on the surface of leaves, which allow gases to move in and out and also let water vapor escape, are called

MEDIUM

- A. Xylem
- B. Phloem
- C. Chloroplast
- D. Stomata (Correct)**

Explanation: Stomata are small pores, usually on the underside of leaves, that allow exchange of gases (like carbon dioxide and oxygen) and release of water vapor during transpiration.

Q32. Which component of human blood is mainly responsible for carrying oxygen to different parts of the body?

MEDIUM

- A. Red blood cells (Correct)**
- B. White blood cells
- C. Platelets
- D. Plasma

Explanation: Red blood cells contain a protein called hemoglobin, which binds to oxygen in the lungs and carries it to tissues throughout the body.

Q33. Which organs mainly filter waste products from the blood and help form urine?

MEDIUM

- A. Small intestine
- B. Kidneys (Correct)**
- C. Lungs
- D. Liver

Explanation: The kidneys filter waste products and excess substances from the blood, forming urine, which is part of the body's excretory system.

Q34. Which cell structure acts as the control center of the cell and contains most of the cell's genetic material?

MEDIUM

- A. Mitochondria
- B. Vacuole
- C. Nucleus (Correct)**
- D. Cell membrane

Explanation: The nucleus contains the cell's genetic material (DNA) and directs the cell's activities, acting as its control center.

Q35. Which organ produces acid that helps break down food during digestion?

MEDIUM

- A. Lungs
- B. Heart
- C. Brain
- D. Stomach (Correct)**

Explanation: The stomach produces gastric acid, which helps break down food and creates conditions suitable for digestive enzymes to act.

Q36. The force that opposes the relative motion between two surfaces in contact is called

MEDIUM

- A. Friction (Correct)**
- B. Gravity
- C. Magnetism
- D. Tension

Explanation: Friction is the force that acts between two surfaces in contact and opposes their relative sliding motion.

Q37. Sound cannot travel through a vacuum (completely empty space) because sound needs

MEDIUM

- A. complete darkness
- B. a material medium to travel through (Correct)**
- C. light to travel alongside it
- D. a magnetic field

Explanation: Sound travels as vibrations passed between particles of a medium (like air, water, or solids); with no particles present in a vacuum, sound cannot travel.

Q38. For electric current to flow continuously through a circuit and light a bulb, the circuit must be

MEDIUM

- A. open at some point
- B. exposed to sunlight
- C. a complete, closed loop (Correct)**
- D. made entirely of insulators

Explanation: Electric current flows only when the circuit forms a complete, closed path from the source, through the components, and back; a break anywhere stops the flow.

Q39. Materials like copper and aluminum, which allow electric current to pass through them easily, are called

MEDIUM

- A. Insulators
- B. Semiconductors only
- C. Magnets
- D. Conductors (Correct)**

Explanation: Conductors are materials, typically metals like copper and aluminum, that allow electric current to flow through them easily.

Q40. If a car covers a certain distance in a certain amount of time, its speed is calculated as

MEDIUM

- A. distance divided by time (Correct)**
- B. time divided by distance
- C. distance multiplied by time
- D. distance minus time

Explanation: Speed is defined as the distance traveled divided by the time taken to travel that distance.

Q41. A ramp used to push a heavy box up to a higher level, instead of lifting it straight up, is an example of which simple machine?

MEDIUM

- A. Gear
- B. Inclined plane (Correct)**
- C. Pulley
- D. Wheel and axle

Explanation: An inclined plane is a flat sloped surface that allows a load to be raised to a height using a smaller force applied over a longer distance, compared to lifting it straight up.

Q42. A substance made of only one type of atom, which cannot be broken down into simpler substances by ordinary chemical methods, is called

MEDIUM

- A. Mixture
- B. Solution
- C. Element (Correct)**
- D. Compound

Explanation: An element is a pure substance made of only one type of atom and cannot be broken down into simpler substances by ordinary chemical reactions, e.g., oxygen or iron.

Q43. Water is a compound formed by the chemical combination of which two elements?

MEDIUM

- A. Carbon and oxygen
- B. Nitrogen and hydrogen
- C. Hydrogen and nitrogen
- D. Hydrogen and oxygen (Correct)**

Explanation: Water is a compound made of hydrogen and oxygen atoms chemically bonded together, commonly represented as H_2O .

Q44. When blue litmus paper is dipped into an acidic solution, it typically turns

MEDIUM

- A. Red (Correct)**
- B. Green
- C. Yellow
- D. Stays blue

Explanation: Acidic solutions turn blue litmus paper red; this color change is a standard test used to identify acids.

Q45. The basic functional unit of all living organisms is the:

MEDIUM

- A. Tissue
- B. Cell (Correct)**
- C. Organ
- D. Nucleus

Explanation: The cell is the smallest structural and functional unit of life; tissues and organs are built from groups of cells.

Q46. A car starts from rest and attains a velocity of 20 m/s in 10 seconds while moving in a straight line. What is its acceleration?

MEDIUM

- A. 200 m/s²
- B. 0.5 m/s²
- C. 2 m/s² (Correct)**
- D. 20 m/s²

Explanation: Acceleration $a = (v - u)/t = (20 - 0)/10 = 2 \text{ m/s}^2$.

Q47. A bullet of mass 20 g is fired from a rifle with a velocity of 100 m/s. What is the momentum of the bullet?

MEDIUM

- A. 5 kg m/s
- B. 20 kg m/s
- C. 2000 kg m/s
- D. 2 kg m/s (Correct)**

Explanation: Momentum $p = mv$. Converting mass to SI units: $20 \text{ g} = 0.02 \text{ kg}$. So $p = 0.02 \text{ kg} \times 100 \text{ m/s} = 2 \text{ kg m/s}$.

Q48. The acceleration due to gravity on the Moon's surface is about 1/6th of that on Earth. If an object weighs 60 N on Earth, what would it weigh on the Moon (approximately)?

MEDIUM

- A. 10 N (Correct)**
- B. 60 N
- C. 360 N
- D. 6 N

Explanation: Weight = mg . Since g on the Moon is about 1/6 of g on Earth, the weight on the Moon = $60/6 = 10 \text{ N}$. The mass stays the same everywhere; only weight changes because it depends on the local value of g .

Q49. A body of mass 2 kg is moving with a velocity of 3 m/s. What is its kinetic energy?

MEDIUM

- A. 6 J
- B. 9 J (Correct)**
- C. 18 J
- D. 3 J

Explanation: $KE = \frac{1}{2}mv^2 = \frac{1}{2} \times 2 \times 3^2 = \frac{1}{2} \times 2 \times 9 = 9 \text{ J}$.

Q50. Sound travels fastest in which of the following media?

MEDIUM

- A. air
- B. water
- C. steel (Correct)**
- D. vacuum

Explanation: Sound travels by the vibration of particles, and it travels fastest in solids like steel because their particles are closely packed, allowing vibrations to transfer quickly. The general order of speed is solids > liquids > gases.

Q51. Why does a solid have a fixed shape and volume, unlike a liquid or a gas?

MEDIUM

- A. because solid particles have maximum kinetic energy
- B. because solid particles have large intermolecular spaces
- C. because solid particles are free to move randomly in all directions
- D. because solid particles have strong intermolecular forces holding them in fixed positions (Correct)**

Explanation: In solids, the intermolecular forces of attraction are very strong, keeping particles tightly packed in fixed positions, which gives solids a definite shape and volume.

Q52. Which cell organelle is mainly responsible for releasing energy from food during cellular respiration, earning it the nickname 'powerhouse of the cell'?

HARD

- A. Nucleus
- B. Mitochondria (Correct)**
- C. Ribosome
- D. Golgi apparatus

Explanation: Mitochondria carry out cellular respiration, breaking down food molecules to release energy that the cell can use, which is why they are called the powerhouse of the cell.

Q53. Besides giving the body its basic shape and support, which of the following is also a key function of the human skeletal system?

HARD

- A. Producing digestive enzymes
- B. Carrying oxygen in the bloodstream
- C. Protecting internal organs, such as the brain and heart (Correct)**
- D. Pumping blood through the body

Explanation: Along with support and movement, the skeletal system protects delicate internal organs — for example, the skull protects the brain and the rib cage protects the heart and lungs.

Q54. Which body system, made up of the brain, spinal cord, and nerves, is mainly responsible for coordinating and controlling body activities by sending and receiving signals?

HARD

- A. Circulatory system
- B. Digestive system
- C. Respiratory system
- D. Nervous system (Correct)**

Explanation: The nervous system, consisting of the brain, spinal cord, and nerves, coordinates body activities by transmitting electrical and chemical signals throughout the body.

Q55. Which of these statements correctly describes a typical difference between arteries and veins in the human body?

HARD

- A. Arteries generally carry blood away from the heart, while veins generally carry blood toward the heart (Correct)**
- B. Arteries generally carry blood toward the heart, while veins carry it away
- C. Only arteries carry blood, veins carry only air
- D. Arteries and veins are two names for the exact same type of blood vessel

Explanation: As a general rule, arteries carry blood away from the heart to the rest of the body, while veins carry blood back toward the heart.

Q56. Chlorophyll, the green pigment in plant leaves, plays which key role in photosynthesis?

HARD

- A. It protects the plant from insects
- B. It absorbs sunlight energy needed to drive the food-making process (Correct)**
- C. It absorbs water from the soil for the plant
- D. It transports food to different parts of the plant

Explanation: Chlorophyll absorbs light energy from the sun, and this energy is used to drive the chemical reactions of photosynthesis that make food for the plant.

Q57. A book resting on a table stays at rest unless a force acts on it, and a rolling ball keeps rolling unless something like friction slows it down. This tendency of objects to resist changes in their state of motion is called

HARD

- A. Momentum only
- B. Acceleration
- C. Inertia (Correct)**
- D. Gravity

Explanation: Inertia is the natural tendency of an object to resist any change in its state of rest or uniform motion, unless acted upon by an external force — this is described by Newton's first law of motion.

Q58. A convex lens, which is thicker in the middle than at its edges, generally makes parallel rays of light

HARD

- A. diverge (spread apart) from each other
- B. stop traveling entirely
- C. reverse their direction completely
- D. converge (come together) toward a point (Correct)**

Explanation: A convex lens bends parallel light rays passing through it so that they converge (meet) at a point called the focus, which is why convex lenses are also called converging lenses.

Q59. The bending of light as it passes from one transparent medium into another (such as from air into water) is called

HARD

- A. Refraction (Correct)**
- B. Reflection
- C. Diffusion
- D. Absorption

Explanation: Refraction is the bending of light when it passes from one transparent medium into another with a different density, such as from air into water, due to a change in its speed.

Q60. In a simple series circuit with multiple bulbs, if one bulb burns out (breaks), what typically happens to the other bulbs in that same series loop?

HARD

- A. They start working in reverse
- B. They also stop working, since the circuit is broken (Correct)**
- C. They continue working exactly as before
- D. They become brighter than before

Explanation: In a series circuit, all components share a single path for current; if one bulb breaks, the path becomes incomplete and current stops flowing, so all bulbs in that series loop stop working.

Q61. An echo is heard when sound waves reflect off a distant surface, such as a wall or a cliff, and return to the listener. This phenomenon is best described as

HARD

- A. Absorption of sound waves by a surface
- B. Generation of brand-new sound waves by the surface
- C. Reflection of sound waves off a surface (Correct)**
- D. Refraction of sound waves through a surface

Explanation: An echo occurs because sound waves bounce back (reflect) off a hard, distant surface and travel back to the listener's ear, similar to how light reflects off a mirror.

Q62. Using a single fixed pulley at the top of a flagpole to raise a flag by pulling the rope downward is mainly useful because it

HARD

- A. eliminates the need for any force entirely
- B. makes the flag heavier so it stays in place
- C. converts the rope into an electrical conductor
- D. changes the direction of the force you need to apply, making it easier and more convenient to lift the flag (Correct)**

Explanation: A single fixed pulley mainly changes the direction of the applied force — pulling down on one side raises the load on the other side — making tasks like raising a flag more convenient, even though it does not reduce the amount of force needed.

Q63. When an acid and a base are mixed together in the right amounts, they react to cancel out each other's properties, forming a salt and water. This reaction is called

HARD

- A. Neutralization (Correct)**
- B. Oxidation
- C. Evaporation
- D. Filtration

Explanation: Neutralization is the chemical reaction between an acid and a base that cancels out their acidic and basic properties, producing a salt and water.

Q64. Rusting of iron, in which a reddish-brown coating forms on iron objects over time, mainly happens when iron reacts with

HARD

- A. Carbon dioxide alone, with no moisture involved
- B. Oxygen and moisture (water) in the air (Correct)**
- C. Nitrogen gas alone
- D. Sunlight alone, with no air or water involved

Explanation: Rusting occurs when iron reacts with oxygen in the presence of moisture (water), forming iron oxide, commonly seen as a reddish-brown coating on iron objects.

Q65. Which of these statements about metals and non-metals is generally correct?

HARD

- A. Metals and non-metals conduct heat and electricity exactly equally well in all cases
- B. Neither metals nor non-metals can conduct electricity under any circumstances
- C. Most metals are good conductors of heat and electricity, while most non-metals are poor conductors (Correct)**
- D. Most metals are poor conductors of electricity, while most non-metals are excellent conductors

Explanation: As a general rule in basic chemistry, most metals (like copper, aluminum, and iron) are good conductors of heat and electricity, whereas most non-metals are poor conductors, with graphite (a form of carbon) being a well-known exception.

Q66. Which of these correctly distinguishes a mixture from a compound?

HARD

- A. A compound can always be separated by physical methods, while a mixture cannot be separated at all
- B. There is no real difference between a mixture and a compound; the terms mean the same thing
- C. A mixture always has a fixed, unchangeable ratio of its components, just like a compound
- D. A mixture can generally be separated into its components by physical methods, while a compound requires a chemical reaction to be broken into its constituent elements (Correct)**

Explanation: A mixture consists of substances that are physically combined and can usually be separated by physical methods (like filtration or evaporation), while a compound consists of elements chemically bonded together in a fixed ratio, requiring a chemical reaction to separate them into their elements.

Q67. A stone is dropped from a height and falls freely under gravity ($g = 10 \text{ m/s}^2$), hitting the ground after 4 seconds. What distance did it fall?

HARD

- A. 40 m
- B. 80 m (Correct)**
- C. 20 m
- D. 160 m

Explanation: Using $s = ut + \frac{1}{2}gt^2$, with $u = 0$: $s = \frac{1}{2} \times 10 \times 4^2 = \frac{1}{2} \times 10 \times 16 = 80 \text{ m}$.

Q68. A force of 15 N acting on a body produces an acceleration of 3 m/s². If the same force acts on a second body of half the mass, what will be the new acceleration?

HARD

- A. 1.5 m/s²
- B. 3 m/s²
- C. 6 m/s² (Correct)**
- D. 12 m/s²

Explanation: From $F = ma$, the original mass $m = F/a = 15/3 = 5$ kg. Half of this mass is 2.5 kg. The new acceleration $= F/m' = 15/2.5 = 6$ m/s². Since acceleration is inversely proportional to mass, halving the mass doubles the acceleration.

Q69. Two objects of masses 4 kg and 9 kg are separated by a distance of 3 m. If G is the universal gravitational constant, what is the gravitational force of attraction between them (in terms of G)?

HARD

- A. 12G
- B. 36G
- C. 0.25G
- D. 4G (Correct)**

Explanation: $F = G(m_1m_2)/r^2 = G(4 \times 9)/3^2 = 36G/9 = 4G$.

Q70. A machine does 500 J of work in 25 seconds. What is the power of the machine?

HARD

- A. 20 W (Correct)**
- B. 12,500 W
- C. 0.05 W
- D. 525 W

Explanation: Power = Work/time = 500/25 = 20 W.

Q71. A person shouts near a cliff and hears the echo after 2 seconds. If the speed of sound in air is 340 m/s, how far is the person from the cliff?

HARD

- A. 680 m
- B. 340 m (Correct)**
- C. 170 m
- D. 1360 m

Explanation: The sound travels to the cliff and back in 2 seconds, so the total distance covered by sound = speed $\times$ time = 340 $\times$ 2 = 680 m. Since this is a round trip (there and back), the distance to the cliff = 680/2 = 340 m.

Q72. Why does the temperature of a substance remain constant during a change of state (for example, while ice is melting at 0°C), even though heat is continuously being supplied?

HARD

- A. because heat supplied is reflected back and not absorbed by the substance
- B. because the substance stops absorbing heat once melting begins
- C. because the supplied heat is used to overcome the intermolecular forces of attraction (latent heat), not to raise the temperature (Correct)**
- D. because the surrounding air absorbs all the heat before it reaches the substance

Explanation: During a change of state, the heat energy supplied (latent heat) is used to weaken and overcome the intermolecular forces holding the particles together, allowing the change of state to occur, rather than increasing the kinetic energy (and hence the temperature) of the particles.

Q73. A mixture of iron filings and sulphur powder needs to be separated. Which property is used to achieve this?

HARD

- A. both are soluble in water, so filtration can separate them
- B. sulphur sublimes on heating, so it can be separated by evaporation
- C. iron filings dissolve in water while sulphur does not
- D. iron filings are magnetic and get attracted by a magnet, while sulphur is not (Correct)**

Explanation: Iron is a magnetic material, so a magnet can be used to attract and pull out the iron filings, leaving behind the non-magnetic sulphur powder — a simple physical separation method based on magnetic properties.

Social Science (73 questions)

Q1. Which river is known as the longest river flowing entirely within India?

EASY

- A. Ganga (Correct)**
- B. Godavari
- C. Yamuna
- D. Brahmaputra

Explanation: The Ganga originates in the Himalayas and flows through several northern Indian states before draining into the Bay of Bengal, making it the longest river in India.

Q2. Which peak, located within India's undisputed territory, is the highest mountain peak in the country?

EASY

- A. Mount Everest
- B. Kangchenjunga (Correct)**
- C. K2
- D. Nanda Devi

Explanation: Kangchenjunga, located in Sikkim on the India-Nepal border, is the highest peak situated within India.

Q3. Which leader of India's freedom struggle is popularly honoured as the 'Father of the Nation'?

EASY

- A. Jawaharlal Nehru
- B. Sardar Vallabhbhai Patel
- C. Mahatma Gandhi (Correct)**
- D. Subhas Chandra Bose

Explanation: Mahatma Gandhi is popularly honoured as the 'Father of the Nation' for his leadership of India's non-violent freedom movement.

Q4. In which year did India become independent from British rule?

EASY

- A. 1942
- B. 1950
- C. 1930
- D. 1947 (Correct)**

Explanation: India became independent from British rule on 15 August 1947.

Q5. What type of government does India have?

EASY

- A. A parliamentary democratic republic (Correct)**
- B. An absolute monarchy
- C. A military dictatorship
- D. A one-party state

Explanation: India is a sovereign, socialist, secular, democratic republic that follows a parliamentary system of government.

Q6. Which of the following is the supreme law of India?

EASY

- A. The Preamble
- B. The Constitution (Correct)**
- C. Acts of Parliament
- D. The Directive Principles of State Policy

Explanation: The Constitution of India is the supreme law of the land, and all other laws must conform to it.

Q7. Materials and substances found in nature, such as water, air, soil and minerals, that can be used by humans are called what?

EASY

- A. Man-made resources
- B. Capital goods
- C. Natural resources (Correct)**
- D. Human resources

Explanation: Resources that are provided by nature without human intervention, such as water, minerals and forests, are called natural resources.

Q8. Which Indian state is traditionally known as the 'Spice Garden of India'?

EASY

- A. Punjab
- B. Rajasthan
- C. Assam
- D. Kerala (Correct)**

Explanation: Kerala is traditionally known as the 'Spice Garden of India' owing to its long history of cultivating pepper, cardamom and other spices.

Q9. In which year was the Quit India Movement launched?

EASY

- A. 1942 (Correct)**
- B. 1930
- C. 1947
- D. 1919

Explanation: The Quit India Movement was launched by Mahatma Gandhi in August 1942, demanding an end to British rule.

Q10. In which part of the Indian Constitution are the Fundamental Rights enumerated?

EASY

- A. Part II
- B. Part III (Correct)**
- C. Part IV
- D. Part V

Explanation: Fundamental Rights are enumerated in Part III of the Indian Constitution.

Q11. Who is regarded as the Father of the Indian Constitution?

EASY

- A. Mahatma Gandhi
- B. Dr. B. R. Ambedkar (Correct)**
- C. Jawaharlal Nehru
- D. Sardar Vallabhbhai Patel

Explanation: Dr. B. R. Ambedkar chaired the Constitution Drafting Committee and is widely regarded as the chief architect of the Indian Constitution.

Q12. In which year did the storming of the Bastille fortress-prison take place, marking a famous event of the French Revolution?

EASY

- A. 1789 (Correct)**
- B. 1776
- C. 1804
- D. 1815

Explanation: The Bastille was stormed by Parisian crowds on 14 July 1789, an event that became a defining symbol of the start of the French Revolution.

Q13. Which assembly, made up of representatives of the clergy, nobility, and common people, did King Louis XVI summon in 1789 to approve new taxes?

EASY

- A. National Convention
- B. Estates General (Correct)**
- C. Directory
- D. Constituent Assembly

Explanation: Louis XVI summoned the Estates General, an assembly of the three estates (clergy, nobility, and commoners), in May 1789 to raise support for new taxes.

Q14. Who was the Tsar (emperor) of Russia at the time of the Russian Revolution of 1917?

EASY

- A. Alexander III
- B. Peter the Great
- C. Nicholas II (Correct)**
- D. Ivan the Terrible

Explanation: Nicholas II was the reigning Tsar of Russia during the 1917 revolution; he abdicated in March 1917 and the Bolsheviks later took power in November 1917.

Q15. Who led the Bolshevik Party that seized power in Russia in November 1917?

EASY

- A. Joseph Stalin
- B. Karl Marx
- C. Leon Trotsky
- D. Vladimir Lenin (Correct)**

Explanation: Vladimir Lenin led the Bolshevik Party during the October (November, by the modern calendar) Revolution of 1917, becoming the head of the new Soviet government.

Q16. In which year did Adolf Hitler become the Chancellor of Germany?

EASY

- A. 1933 (Correct)**
- B. 1919
- C. 1939
- D. 1945

Explanation: Adolf Hitler was appointed Chancellor of Germany on 30 January 1933, after which the Nazi Party rapidly dismantled democratic institutions.

Q17. Which system of government is defined as one where rulers are elected by the people and are accountable to them?

EASY

- A. Monarchy
- B. Democracy (Correct)**
- C. Theocracy
- D. Military rule

Explanation: Democracy is a form of government in which the rulers are elected by the people and remain accountable to them through regular elections.

Q18. Who was the Chairman of the Drafting Committee of the Indian Constituent Assembly?

EASY

- A. Jawaharlal Nehru
- B. Rajendra Prasad
- C. B. R. Ambedkar (Correct)**
- D. Sardar Vallabhbhai Patel

Explanation: Dr. B. R. Ambedkar chaired the Drafting Committee of the Constituent Assembly, which was responsible for preparing the draft of the Indian Constitution.

Q19. The electoral system used in most Indian elections, where the candidate with the highest number of votes in a constituency wins, is known by what name?

EASY

- A. Proportional representation
- B. Indirect election system
- C. Single transferable vote system
- D. First-past-the-post system (Correct)**

Explanation: India uses the first-past-the-post system for Lok Sabha and Vidhan Sabha elections, where the candidate securing the most votes in a constituency is declared the winner, even without an absolute majority.

Q20. Which mountain range forms India's northern boundary and separates it from the Tibetan plateau?

EASY

- A. The Himalayas (Correct)**
- B. The Aravallis
- C. The Western Ghats
- D. The Vindhyas

Explanation: The Himalayas, a young and structurally fold mountain range, stretch along India's northern border and separate the Indian subcontinent from the Tibetan plateau.

Q21. Which season, associated with heavy rainfall over most of India, is brought by the south-west monsoon winds?

EASY

- A. Winter season
- B. Rainy (monsoon) season (Correct)**
- C. Summer season
- D. Season of retreating monsoon

Explanation: The south-west monsoon winds, blowing from June to September, bring heavy rainfall to most parts of India during the rainy (monsoon) season.

Q22. Which of the following best describes a defining feature of democracy?

EASY

- A. Only wealthy citizens are allowed to vote
- B. One political party controls the government permanently to ensure stability
- C. Rulers elected by the people are the final decision-makers on major issues (Correct)**
- D. Only the military can decide who becomes the head of government

Explanation: A core feature of democracy, as defined in NCERT Class 9 Civics, is that the final decision-making power rests with representatives elected by the people through free and fair elections, held at regular intervals.

Q23. The NCERT chapter on Constitutional Design begins with the story of the struggle against apartheid in which country?

EASY

- A. South Africa (Correct)**
- B. Russia
- C. Germany
- D. Sri Lanka

Explanation: NCERT Class 9 Civics' chapter on Constitutional Design opens with South Africa's long struggle against apartheid, a system of institutionalised racial discrimination, which ended when Nelson Mandela became the country's first democratically elected president in 1994 and a new, widely-praised constitution was adopted.

Q24. How many countries share a land border with India?

EASY

- A. Five
- B. Nine
- C. Seven (Correct)**
- D. Six

Explanation: India shares its land border with seven countries: Pakistan and Afghanistan to the northwest, China, Nepal, and Bhutan to the north, and Myanmar and Bangladesh to the east, as covered in NCERT Class 9 Geography's chapter on India's size and location.

Q25. In which year did Adolf Hitler become the Chancellor of Germany?

EASY

- A. 1919
- B. 1929
- C. 1939
- D. 1933 (Correct)**

Explanation: Adolf Hitler was appointed Chancellor of Germany on 30 January 1933. He quickly moved to dismantle democratic institutions, and within months, following the Enabling Act, established a one-party Nazi dictatorship.

Q26. Which constitutional body is responsible for conducting free and fair elections in India?

EASY

- A. NITI Aayog
- B. Finance Commission
- C. Election Commission of India (Correct)**
- D. Supreme Court of India

Explanation: The Election Commission of India is an independent constitutional body responsible for conducting free and fair elections to the Lok Sabha, Rajya Sabha, state legislatures, and the offices of President and Vice-President, at the national and state levels.

Q27. Project Tiger, launched to conserve India's tiger population, began in which year?

EASY

- A. 1952
- B. 1973 (Correct)**
- C. 1991
- D. 2002

Explanation: Project Tiger, a wildlife conservation programme launched by the Government of India to protect the endangered tiger, began in 1973, with Jim Corbett National Park among the first reserves brought under it, as covered in NCERT Class 9 Geography's chapter on Natural Vegetation and Wildlife.

Q28. The term 'monsoon', central to India's climate, comes from an Arabic word meaning?

EASY

- A. Rain
- B. Wind
- C. Season (Correct)**
- D. Cloud

Explanation: The word 'monsoon' is derived from the Arabic word 'mausim', meaning 'season', reflecting how India's climate is dominated by seasonally reversing wind patterns, as explained in NCERT Class 9 Geography's chapter on climate.

Q29. In India's parliamentary system of government, who exercises the real executive power?

EASY

- A. The President alone
- B. The Chief Justice of India
- C. The Speaker of the Lok Sabha
- D. The Prime Minister and the Council of Ministers (Correct)**

Explanation: India follows a parliamentary system in which the President is the nominal (constitutional) head of state, while real executive power is exercised by the Prime Minister and the Council of Ministers, who are collectively responsible to the Lok Sabha.

Q30. Which river is popularly called the 'Sorrow of Bihar' because of its history of devastating floods?

MEDIUM

- A. Damodar
- B. Mahanadi
- C. Kosi (Correct)**
- D. Tapi

Explanation: The Kosi river is called the 'Sorrow of Bihar' because it frequently changes its course and has caused devastating floods in the state.

Q31. The Deccan Plateau is mainly located in which part of India?

MEDIUM

- A. Northern India
- B. North-eastern India
- C. A narrow western coastal strip only
- D. Peninsular (southern) India (Correct)**

Explanation: The Deccan Plateau is a large plateau covering much of peninsular (southern) India, bounded by the Western and Eastern Ghats.

Q32. The Thar Desert lies mainly in which Indian state?

MEDIUM

- A. Rajasthan (Correct)**
- B. Gujarat
- C. Haryana
- D. Punjab

Explanation: The Thar Desert, also known as the Great Indian Desert, lies mainly in the state of Rajasthan.

Q33. In which year was the Indian National Congress founded?

MEDIUM

- A. 1857
- B. 1885 (Correct)**
- C. 1905
- D. 1920

Explanation: The Indian National Congress was founded in 1885, becoming one of the earliest and most prominent platforms for organized nationalist activity.

Q34. In which year was the Battle of Plassey fought?

MEDIUM

- A. 1764
- B. 1857
- C. 1757 (Correct)**
- D. 1526

Explanation: The Battle of Plassey, fought in 1757, marked the East India Company's decisive victory that began its political control over Bengal.

Q35. Who chaired the Drafting Committee of the Constituent Assembly responsible for drafting the Constitution of India?

MEDIUM

- A. Jawaharlal Nehru
- B. Rajendra Prasad
- C. Sardar Patel
- D. Dr. B.R. Ambedkar (Correct)**

Explanation: Dr. B.R. Ambedkar chaired the Drafting Committee of the Constituent Assembly responsible for drafting the Constitution of India.

Q36. The Fundamental Duties were added to the Indian Constitution by which constitutional amendment?

MEDIUM

- A. 42nd Amendment (Correct)**
- B. 44th Amendment
- C. 61st Amendment
- D. 73rd Amendment

Explanation: The Fundamental Duties were added to the Constitution through the 42nd Amendment Act of 1976.

Q37. Who heads the Supreme Court and, by convention, the Indian judiciary?

MEDIUM

- A. President of India
- B. Chief Justice of India (Correct)**
- C. Attorney General of India
- D. Speaker of the Lok Sabha

Explanation: The Chief Justice of India heads the Supreme Court and, by convention, the Indian judiciary.

Q38. What is the minimum age required to contest election as a member of the Lok Sabha?

MEDIUM

- A. 18 years
- B. 21 years
- C. 25 years (Correct)**
- D. 30 years

Explanation: A person must be at least 25 years old to be eligible to contest election as a member of the Lok Sabha.

Q39. Which sector of the economy includes activities that directly use natural resources, such as agriculture, fishing and mining?

MEDIUM

- A. Secondary sector
- B. Tertiary sector
- C. Quaternary sector
- D. Primary sector (Correct)**

Explanation: The primary sector includes activities that directly use natural resources, such as agriculture, fishing and mining.

Q40. What does a 'barter system' of trade mean?

MEDIUM

- A. Exchange of goods for goods without the use of money (Correct)**
- B. Buying goods using credit cards
- C. Trading only agricultural goods
- D. Government-controlled trade of goods

Explanation: A barter system is a method of trade in which goods or services are exchanged directly for other goods or services, without the use of money.

Q41. Which Indian state is the largest by geographical area?

MEDIUM

- A. Madhya Pradesh
- B. Rajasthan (Correct)**
- C. Maharashtra
- D. Uttar Pradesh

Explanation: Rajasthan is the largest Indian state by geographical area.

Q42. On which date did the Constituent Assembly formally adopt the Constitution of India?

MEDIUM

- A. 15 August 1947
- B. 26 January 1950
- C. 26 November 1949 (Correct)**
- D. 2 October 1948

Explanation: The Constituent Assembly formally adopted the Constitution of India on 26 November 1949, a date now observed as Constitution Day.

Q43. Who led the Dandi (Salt) March in 1930 to protest the British salt tax?

MEDIUM

- A. Jawaharlal Nehru, in 1930
- B. Mahatma Gandhi, in 1942
- C. Sardar Patel, in 1930
- D. Mahatma Gandhi, in 1930 (Correct)**

Explanation: Mahatma Gandhi led the Dandi (Salt) March in 1930 to protest the British salt tax as part of the civil disobedience movement.

Q44. Which mountain range forms a natural boundary along much of India's northern frontier, including areas bordering China?

MEDIUM

- A. Himalayas (Correct)**
- B. Vindhya Range
- C. Western Ghats
- D. Aravalli Range

Explanation: The Himalayan mountain range forms a natural boundary along much of India's northern frontier, including areas bordering China.

Q45. The Tropic of Cancer does NOT pass through which of these Indian states?

MEDIUM

- A. Gujarat
- B. Madhya Pradesh
- C. Punjab (Correct)**
- D. West Bengal

Explanation: The Tropic of Cancer passes through 8 Indian states including Gujarat, Madhya Pradesh, and West Bengal, but not Punjab, which lies further north.

Q46. The 'Declaration of the Rights of Man and of the Citizen', adopted in 1789, was primarily inspired by which political idea?

MEDIUM

- A. Divine right of kings to rule
- B. Superiority of the clergy over common citizens
- C. Natural and equal rights of all individuals (Correct)**
- D. Exclusive political rights for the nobility

Explanation: The Declaration of the Rights of Man and of the Citizen enshrined the idea that all men are born free and with equal natural rights, drawing on Enlightenment philosophy.

Q47. Which key feature of India's electoral system gives every citizen aged 18 or above the right to vote, regardless of caste, religion, gender, or property ownership?

MEDIUM

- A. Reservation of constituencies for a fixed group of candidates
- B. Nomination of members by the President
- C. Property-based qualification for voting
- D. Universal adult franchise (Correct)**

Explanation: Universal adult franchise, guaranteed under the Indian Constitution, gives every citizen aged 18 and above the right to vote in elections without any qualification based on caste, religion, gender, education, or property, making it a core feature of India's electoral democracy.

Q48. Which feature of the Indian Constitution allows a citizen to move the Supreme Court or High Courts directly if any of their Fundamental Rights is violated?

MEDIUM

- A. The justiciable nature of Fundamental Rights (Correct)**
- B. The Directive Principles of State Policy
- C. The ideals stated in the Preamble alone
- D. The administrative provisions of the Fifth Schedule

Explanation: Fundamental Rights, listed in Part III of the Constitution, are justiciable — meaning courts can directly enforce them — so a citizen whose Fundamental Right is violated can approach the Supreme Court (Article 32) or a High Court (Article 226) for a remedy.

Q49. What was a key demand of the workers and soldiers who supported the Bolsheviks in the events leading to the November 1917 revolution in Russia?

MEDIUM

- A. Restoration of absolute power to the Tsar
- B. An end to Russia's participation in World War I and redistribution of land (Correct)**
- C. Expansion of the Russian empire into new colonies
- D. Strengthening the powers of the landed nobility

Explanation: Widespread war-weariness, food shortages, and peasant demands for land reform fuelled support for the Bolsheviks, who promised 'Peace, Land, and Bread' — withdrawing from World War I and redistributing land to peasants.

Q50. The Weimar Republic, established in Germany after World War I, faced which major challenge that weakened public faith in democracy before Hitler's rise?

MEDIUM

- A. A sudden abundance of gold reserves
- B. Total absence of any political parties
- C. Severe hyperinflation and economic depression (Correct)**
- D. Complete freedom from the Treaty of Versailles' obligations

Explanation: The Weimar Republic suffered from severe hyperinflation in the early 1920s and then the Great Depression from 1929, which caused mass unemployment and eroded public confidence in the democratic government.

Q51. What was the Holocaust, carried out under Nazi rule during World War II?

MEDIUM

- A. A programme to improve public healthcare across Europe
- B. An economic reform plan to reduce German unemployment
- C. A treaty ending hostilities between Germany and its neighbours
- D. The systematic, state-organised genocide of six million Jews and other targeted groups (Correct)**

Explanation: The Holocaust refers to the systematic, state-sponsored persecution and mass murder of approximately six million Jews, along with Roma people, disabled persons, and other groups, carried out by the Nazi regime.

Q52. In which landmark case did the Supreme Court of India propound the 'basic structure' doctrine of the Constitution?

HARD

- A. Golaknath case
- B. Kesavananda Bharati case (Correct)**
- C. Minerva Mills case
- D. Shah Bano case

Explanation: The Kesavananda Bharati case (1973) is the landmark Supreme Court judgment in which the 'basic structure' doctrine of the Constitution was propounded.

Q53. Which Article of the Indian Constitution abolishes untouchability?

HARD

- A. Article 14
- B. Article 15
- C. Article 17 (Correct)**
- D. Article 21

Explanation: Article 17 of the Constitution abolishes untouchability and forbids its practice in any form.

Q54. Which Schedule of the Indian Constitution contains provisions relating to disqualification of legislators on grounds of defection?

HARD

- A. Ninth Schedule
- B. Eleventh Schedule
- C. Twelfth Schedule
- D. Tenth Schedule (Correct)**

Explanation: The Tenth Schedule of the Constitution, added by the 52nd Amendment, contains provisions relating to disqualification of legislators on grounds of defection.

Q55. Who served as the first Governor-General of independent India after 15 August 1947?

HARD

- A. Lord Mountbatten (Correct)**
- B. Lord Curzon
- C. C. Rajagopalachari
- D. Lord Wavell

Explanation: Lord Mountbatten served as the first Governor-General of independent India after 15 August 1947.

Q56. Which leader organized and led the Indian National Army (INA), also known as the Azad Hind Fauj?

HARD

- A. Bhagat Singh
- B. Subhas Chandra Bose (Correct)**
- C. Lala Lajpat Rai
- D. Bal Gangadhar Tilak

Explanation: Subhas Chandra Bose organized and led the Indian National Army (INA), also known as the Azad Hind Fauj, to fight for India's independence.

Q57. Which river originates from the Amarkantak plateau in Madhya Pradesh and flows westward into the Arabian Sea?

HARD

- A. Godavari
- B. Krishna
- C. Narmada (Correct)**
- D. Tapi

Explanation: The Narmada river originates from the Amarkantak plateau in Madhya Pradesh and flows westward into the Arabian Sea.

Q58. The Tropic of Cancer passes through how many Indian states?

HARD

- A. Six states
- B. Seven states
- C. Nine states
- D. Eight states (Correct)**

Explanation: The Tropic of Cancer passes through eight Indian states, including Gujarat, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand, West Bengal, Tripura and Mizoram.

Q59. Which sector of the economy comprises services such as banking, transport, communication and trade?

HARD

- A. Tertiary sector (Correct)**
- B. Manufacturing (secondary) sector
- C. Primary sector
- D. Agricultural sector

Explanation: The tertiary sector comprises services such as banking, transport, communication and trade.

Q60. What term describes goods and services purchased from other countries and brought into one's own country?

HARD

- A. Exports
- B. Imports (Correct)**
- C. Subsidies
- D. Tariffs

Explanation: Goods and services purchased from other countries and brought into one's own country are called imports.

Q61. Which Article of the Constitution provides the Right to Constitutional Remedies, allowing citizens to approach the Supreme Court directly for enforcement of Fundamental Rights?

HARD

- A. Article 19
- B. Article 21
- C. Article 32 (Correct)**
- D. Article 25

Explanation: Article 32 provides the Right to Constitutional Remedies, allowing citizens to approach the Supreme Court directly for enforcement of Fundamental Rights.

Q62. Who was the first President of India, holding office from 1950?

HARD

- A. Dr. S. Radhakrishnan
- B. Zakir Hussain
- C. Jawaharlal Nehru
- D. Dr. Rajendra Prasad (Correct)**

Explanation: Dr. Rajendra Prasad served as the first President of India, holding office from 1950.

Q63. Many structural and federal features of the Indian Constitution, such as the scheme of provincial autonomy, were drawn from which earlier legislation?

HARD

- A. Government of India Act, 1935 (Correct)**
- B. Regulating Act, 1773
- C. Indian Councils Act, 1909
- D. Charter Act, 1833

Explanation: Many structural and federal features of the Indian Constitution, such as the scheme of provincial autonomy and the federal framework, were drawn from the Government of India Act, 1935.

Q64. Which Indian state has the longest coastline?

HARD

- A. Tamil Nadu
- B. Gujarat (Correct)**
- C. Andhra Pradesh
- D. Maharashtra

Explanation: Gujarat has the longest coastline among Indian states.

Q65. The words 'socialist', 'secular' and 'integrity' were added to the Preamble of the Constitution by which amendment?

HARD

- A. 24th Amendment
- B. 61st Amendment
- C. 42nd Amendment (Correct)**
- D. 86th Amendment

Explanation: The 42nd Amendment Act of 1976 added the words 'socialist', 'secular' and 'integrity' to the Preamble of the Constitution.

Q66. Resources that are created by humans using natural resources, such as buildings, machinery and roads, are called what?

HARD

- A. Natural resources
- B. Renewable resources
- C. Non-renewable resources
- D. Human-made resources (Correct)**

Explanation: Resources that are created by humans by transforming or utilizing natural resources, such as buildings, machinery and roads, are called human-made resources.

Q67. Which of the following is a well-known criticism of India's first-past-the-post electoral system?

HARD

- A. It requires every voter to rank all candidates in order of preference
- B. A party can win a large majority of seats in the legislature while receiving less than half of the total votes cast across the country (Correct)**
- C. It guarantees each party a share of seats exactly equal to its share of votes
- D. It is used only for elections to the Rajya Sabha and not for the Lok Sabha or state assemblies

Explanation: Under first-past-the-post, the candidate with the highest number of votes in a constituency wins, even without a majority; when a party wins many constituencies by narrow margins, its share of seats nationwide can be much higher than its share of the total votes — a frequently cited disproportionality of this system.

Q68. The Jacobins, under whose influence France experienced the 'Reign of Terror' (1793-94), were most associated with which of these positions?

HARD

- A. They represented the wealthiest sections of society and sought to restore the monarchy's absolute power
- B. They were loyal supporters of Louis XVI who opposed any change to the constitution
- C. They represented poorer and radical sections, favoured a republic, and used strict measures against those seen as enemies of the revolution (Correct)**
- D. They were a foreign army sent to suppress the French Revolution

Explanation: The Jacobins, led by figures such as Maximilien Robespierre, represented poorer, radical sections of society, pushed for an end to monarchy and a republic, and directed strict measures — including mass executions by guillotine — against those they considered enemies of the revolution during the Reign of Terror.

Q69. In the lead-up to the 1917 February Revolution in Russia, which factor most directly triggered the mass protests in Petrograd that forced the Tsar's abdication?

HARD

- A. A sudden and unexpected agreement between the Tsar and the Bolsheviks
- B. A decisive Russian military victory over Germany
- C. A nationwide referendum in which citizens voted to end the monarchy
- D. Severe food and bread shortages combined with war-weariness among soldiers and workers (Correct)**

Explanation: By February 1917, food and bread shortages in Petrograd, combined with the exhaustion of soldiers and workers from Russia's costly involvement in World War I, triggered strikes and demonstrations that escalated into a revolution, leading to Tsar Nicholas II's abdication in March 1917.

Q70. Which best describes the difference between the February and October (November) Revolutions of 1917 in Russia?

HARD

- A. The February Revolution overthrew the Tsar and established a Provisional Government, while the October Revolution saw the Bolsheviks overthrow that Provisional Government (Correct)**
- B. Both revolutions happened on the exact same day and achieved identical outcomes
- C. The February Revolution was led by the Bolsheviks, while the October Revolution restored the Tsar to power
- D. The October Revolution occurred decades before the February Revolution

Explanation: The February Revolution (March 1917 by the modern calendar) forced Tsar Nicholas II's abdication and led to a Provisional Government; the October Revolution (November 1917) was led by the Bolsheviks under Lenin, who overthrew this Provisional Government and established Soviet rule.

Q71. Which of these factors most directly helped the Nazi Party grow from a fringe group to Germany's largest party by 1932?

HARD

- A. A landslide victory in a free and fair national referendum held in 1920
- B. The Great Depression's mass unemployment, which the Nazis exploited through propaganda blaming Jews, Communists, and the Weimar government (Correct)**
- C. Strong support from the League of Nations for Nazi policies
- D. Germany's rapid economic prosperity throughout the 1920s and 1930s

Explanation: The Great Depression from 1929 caused mass unemployment and economic desperation in Germany; the Nazi Party's propaganda machine, led by figures like Joseph Goebbels, exploited this crisis by scapegoating Jews, Communists, and the Weimar Republic, rapidly expanding Nazi electoral support by 1932.

Q72. In India's parliamentary system, the principle of 'collective responsibility' means the entire Council of Ministers is answerable as a group to which body?

HARD

- A. The President
- B. The Rajya Sabha
- C. The Lok Sabha (Correct)**
- D. The Supreme Court

Explanation: Under collective responsibility, all ministers share responsibility for the government's decisions and must resign together if the Lok Sabha passes a no-confidence motion against the government, since it is the directly elected house whose confidence sustains the government.

Q73. In India's Parliament, what is the primary constitutional distinction in law-making power between the Lok Sabha and the Rajya Sabha regarding Money Bills?

HARD

- A. A Money Bill can only be introduced in the Rajya Sabha and requires Lok Sabha's approval afterward
- B. Both Houses have completely equal and identical powers over Money Bills
- C. Money Bills must be approved by the Supreme Court before being introduced in either House
- D. A Money Bill can only be introduced in the Lok Sabha, and the Rajya Sabha cannot reject or amend it, only delay it briefly and offer recommendations (Correct)**

Explanation: Under the Indian Constitution, a Money Bill can only be introduced in the Lok Sabha; the Rajya Sabha can only make recommendations, which the Lok Sabha may accept or reject, and it must return the bill within 14 days, reflecting the Lok Sabha's greater power in financial matters.